

Photosystem II Function under Extreme Low Light and Low Temperatures; Polar vs. Temperate Phytoplankton

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2026-01-09

Abstract

Polar phytoplankton achieve slow, but ecologically significant, productivity under exceptionally low light, where the sequential arrivals of photons at Photosystem II complexes are widely spaced in time. We hypothesized that maintaining photosynthesis under such low growth light involves suppression of energetically wasteful photochemical ‘misses’ in Photosystem II. Under full darkness Photosystem II relaxes predominantly to the S1 oxidation state of the oxygen evolving complex. After such dark relaxation we applied series of single turnover, saturating, flashes to advance Photosystem II through sequential photochemical charge separations, progressively oxidizing the oxygen evolving complex through sequential increments through the S-States, resulting in four-step changes in Photosystem II chlorophyll fluorescence yield. We tracked these oscillations in chlorophyll fluorescence to measure the persistence of synchronized S-States across the Photosystem II populations in diatoms and green algae, in response to changing measurement temperatures and extremely low effective light levels, which we emulated by increasing the spacing between photon deliveries. Prolonged synchronous S-State cycling indicates fewer desynchronizations from wasteful photochemical misses and, thus, potentially more efficient photosynthetic energy conversion under low growth light. We observed that colder temperatures result in more sustained cycling at longer photon spacings, within taxa. Further, polar taxa maintained synchronized cycles for longer than did temperate taxa, even under comparable measurement conditions. Our findings support our hypothesis that diverse lineages of polar phytoplankton can suppress energetically wasteful photochemical misses under low temperature, to help sustain photosynthesis even under extremely low growth light. Therefore, ambient temperature interacts with the lower-light limitation on photosynthesis.

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Introduction

Phytoplankton are a functional grouping of photosynthetic microorganisms, of diverse lineages and ecologies (Pierella Karlusich et al., 2020). Phytoplankters share

photolithotrophic growth, using a set of highly conserved protein complexes (Falkowski et al., 2008) to harness energy from Photosynthetically Active Radiation (PAR), to drive reduction and assimilation of inorganic carbon, nitrogen, phosphorus, sulphur, and other essential micronutrients (Raven et al., 2000). Phytoplankton taxa span diverse metabolic and regulatory responses (Croteau et al., 2024; Ananyev et al., 2017; Zournas et al., 2023), but all rely upon PAR, which decreases with water depth, as light passing through the water column is scattered and absorbed (Kirk, 2011). Therefore, with photons as an energy input, phytoplankton growth is constrained to the photic zone, the region of water receiving sufficient light for photosynthesis, operationally defined in past as extending down to a limit receiving 1% of surface irradiance, or roughly 2-20 $\mu\text{mol photons m}^{-2} \text{ s}^{-1}$ (Raven et al., 2000; Falkowski, 1994).

Light availability is further constrained in polar aquatic habitats, with light penetration of the water limited by seasonally low solar angles, ice cover, and snow cover (Cohen et al., 2020; Leu et al., 2015; Hancke et al., 2018), thereby presenting challenges for phytoplankton growth (Hancke et al., 2018; Randelhoff et al., 2020; Hoppe et al., 2024). Indeed, the lower limit of the photic zone was reconsidered when benthic microalgae in the Antarctic were reported as photosynthetically active at light levels less than 1 $\mu\text{mol photons m}^{-2} \text{ s}^{-1}$ (Robinson et al., 1995; McMinn et al., 2003). Some psychrophilic phytoplankton further demonstrate slow, but ecologically significant, photosynthetic productivity during the polar winter at PAR below 0.17 $\mu\text{mol photons m}^{-2} \text{ s}^{-1}$ (Randelhoff et al., 2020; Hancke et al., 2018; Morgan-Kiss et al., 2016; Hoppe et al., 2024), with implications for the responses of polar phytoplankton communities to climate change (Ardyna and Arrigo, 2020).

This slow, but potentially significant, phytoplankton photosynthesis during polar winters underscores the ability of psychrophilic phytoplankton to maintain functional photosystems throughout the polar night (Morgan-Kiss et al., 2016; Kvernvik et al., 2018; Randelhoff et al., 2020; Hoppe et al., 2024), and supports a theoretical minimum light level for phytoplankton photosynthesis of $0.01 \mu\text{mol photons m}^{-2} \text{ s}^{-1}$ (Raven et al., 2000). Such low light phytoplankton photosynthesis could mitigate cell mortality in the extended darkness of winter, maintaining seeding populations for the spring bloom (Randelhoff et al., 2020). These spring blooms, in turn, are a major source of annual net primary production in polar regions (Ardyna and Arrigo, 2020).

In parallel with seasonally low light, psychrophile phytoplankton contend with cold temperature constraints on water viscosity, solute diffusion rates, membrane fluidity, enzyme kinetics and macromolecule interactions (Lyon and Mock, 2014).

Salinities vary across polar habitats (Morgan-Kiss et al., 2016; Smoot and Hopcroft, 2017; Hoppmann et al., 2022; Li et al., 2025), and can fluctuate annually, driven by freezing and thawing of ice (Carton et al., 2015). Such high, or variable, salinities further impact photosynthesis. The polar green algae *Chlamydomonas priscui* and *Chlamydomonas* ICE-MDV are naturally found in Antarctic saline waters, where acclimation to high salinity leads to the formation of stable PSI supercomplexes associated with increased rates of cyclic electron flow (CEF) and a higher capacity for nonphotochemical quenching (NPQ), predicted to have a photoprotective role in their natural environment (Kalra et al., 2020; Kalra et al., 2023; Stahl-Rommel et al., 2022). This adaptation may be independent of low temperature, as long term exposure to high salinity also led to restructuring of PSI

supercomplexes and increased rates of CEF and NPQ in *Chlamydomonas reinhardtii*, a temperate species naturally found in low salinities (Kalra et al., 2023).

Such extreme abiotic constraints probably act as powerful ecological filters, by imposing niche-driven speciation processes among phytoplankton lineages adapted to polar environments (Ibarbalz et al., 2023). Some psychrophilic phytoplankton indeed exhibit high genetic divergence from related temperate species. The polar diatom *Fragilariopsis cylindrus* shows divergent alleles compared to their mesophilic relatives (Mock et al., 2017), with changes in protein structure, including amino acid substitutions, H-bonds, and salt bridges (Lyon and Mock, 2014), cold shock proteins (Lyon and Mock, 2014) and anti-freeze proteins (Bayer-Giraldi et al., 2011). Similarly, the polar green algae *Chlamydomonas priscui* and *Chlamydomonas* sp. ICE-L harbour large genomes enriched in genes families encoding for ice-binding proteins, molecular chaperones, light harvesting proteins and core histones (Zhang et al., 2020). Polar microbes use cellular-compatible solutes, including sugars, polyols, amino acids, betaine, and DMSP, which lower intracellular freezing points and maintain enzyme hydration spheres, stabilizing catalytic activity (Lyon and Mock, 2014; Zhang et al., 2020). Additionally, they exhibit high levels of polyunsaturated fatty acids (PUFAs) in their lipid membranes (Lyon and Mock, 2014; Cvetkovska et al., 2022) contributing to maintenance of membrane fluidity at cold temperatures.

Oxygenic photosynthesis in eukaryotic phytoplankton occurs in chloroplasts, bounded by a two- to four-membrane envelope, depending upon taxa (Kirk, 2011). Thylakoid membranes, containing pigments and electron carriers, traverse the chloroplast

stroma. Photosystem II (PSII), a multi-subunit protein complex embedded in the thylakoid membranes (Shevela et al., 2023), catalyzes the photooxidation of water, releasing O_2 and H^+ , and passing reductant into the photosynthetic electron transport chain.

Photons are captured by absorbed by diverse light harvesting complexes, which vary across taxa (Kirk, 2011), initiating a transition from the ground state to an electronically excited state. Energy transfers among multiple pigments before eventually reaching the photochemical reaction center of PSII, P_{680} (Kirk, 2011), composed of a Chl a heterodimer (Shen, 2021; Vass, 2011). The excitation energy is then distributed, in variable proportions, to i) photochemistry; ii) dissipation as heat; or iii) re-emission as fluorescence (ChlF) (Schuback et al., 2021). When P_{680} is raised to its excited state, P_{680}^* , it shifts to a stronger (more negative) reduction potential, allowing photooxidation to P_{680}^+ through transfer of an electron to a phaeophytin (Phe) initial acceptor molecule (Vass, 2011; Shevela et al., 2023). In productive electron transport the electron from reduced Phe $^-$ is transferred to plastoquinone A (Q_A), followed by transfer to plastoquinone B (Q_B) (Shen, 2021). Q_B , once fully reduced after receiving two electrons, is released, carrying the electrons into the mobile plastoquinone pool in the lipid phase of the thylakoid membrane (Kirk, 2011). P_{680}^+ is reduced back to P_{680} by taking an electron from a tyrosine residue D1-Tyr-161 (Yz). Yz, in turn, extracts an electron from a manganese-oxo cluster (Mn_4CaO_5) on the thylakoid lumenal side of PSII, where water is oxidized (Shen, 2021; Mukhopadhyay et al., 2004; Gates et al., 2020). Four consecutive charge separations at P_{680} induce four increasingly oxidized states of the Oxygen Evolving Complex (OEC) Mn cluster, known as S-States (Zaharieva and Dau, 2019), denoted, from most reduced to most oxidized, as S0, S1, S2, and S3, followed by a transient S4 state, which rapidly decays to S0. Once the OEC has

progressively lost four electrons, it oxidizes two water molecules, releasing one molecule of O₂. Therefore, a complete water oxidation cycle during oxygenic photosynthesis requires four sequential photon excitations of a single PSII, with the progressive accumulation of four oxidizing equivalents in the OEC (Dau and Haumann, 2007; Gates et al., 2020; Zaharieva and Dau, 2019), and the reduction of two Q_B which enter linear electron transport.

During photosynthesis, electron transfers thus sequentially stabilize the photochemical charge separations (Vass, 2011). These charge separations are, however, reversible through recombination reactions (Vass and Cser, 2009; Han et al., 2012), which are futile cycles of photochemistry (Rappaport et al., 2002; Rappaport et al., 2005; Laloi et al., 2006) forming one category of photochemical ‘miss’ (Han et al., 2012, 2022). Recombinations can generate chlorophyll triplets which react with ground-state molecular oxygen to produce singlet oxygen (¹O₂), a highly damaging, photoinhibitory reactive oxygen species (ROS) (Rappaport et al., 2005). Conversely, harmless non-radiative charge recombinations act as a mechanism of photoprotection under excess excitation (Vass, 2011), enabling direct recombination from the singlet P₆₈₀+Phe⁻ or P₆₈₀+Q_A⁻ states (Rutherford et al., 2012), to help suppress the formation of harmful ROS (Vass and Cser, 2009). Since these recombinations have characteristic activation energies (Rappaport et al., 2002), the probabilities of recombinations vary with temperature, (Ivanov et al., 2006; Han et al., 2008; Han et al., 2012, 2022). Both plants and cyanobacteria furthermore show acclimatory modulations (Quigg et al., 2003) of the reduction potentials of electron transfer intermediates, thereby changing the probabilities and temperature dependencies of recombinations (Ivanov et al., 2006; Sane et al., 2003; Sane et al., 2002; Ivanov et al., 2003).

Beyond their roles in photodamage and photoprotection, charge recombination reactions are wasteful a category of the 'miss' processes that compete with linear electron transport (Rappaport et al., 2005; Han et al., 2012; Ananyev et al., 2017; Zournas et al., 2023). Shifting reduction potentials of downstream electron acceptors, leading to changes in energy gaps for recombinations, may represent evolutionary adaptations aimed at maximizing photoprotection and minimizing losses through back-reactions under light-limited conditions (Vinyard et al., 2013).

Under light limitation, the excitations of P_{680} in each PSII are widely spaced in time. Thus, in order to maintain productive electron transport, the electron transfer intermediates must persist for long periods of time, elevating the probabilities of energetically wasteful recombination reactions, which represent a step backward in the S-State cycle (Keren et al., 1997; de Wijn and van Gorkom, 2002; Han et al., 2008). Furthermore, the probabilities of charge recombinations decrease as temperature drops below the activation energies for recombination paths, 21 - 29 °C for recombination from Q_A^- to S2, or 35 - 41 °C for recombination from Q_B^- to S2 (Ivanov et al., 2006). Stable and progressive S-State cycling under low light, in contrast, sustains electron flow for ATP and NADPH production, while minimizing risks of low-light photodamage to PSII (Rappaport et al., 2005; Keren et al., 1997).

S-State cycling in a phytoplankton sample can be tracked by applications of sequences of short, single-turnover saturating light flashes, bright enough to deliver a photon to each PSII. In darkness the PSII population relaxes primarily to S1 (Han et al., 2008). As sequential light flashes are applied, the population of PSII complexes is driven

through the S-State cycle (Dau and Haumann, 2007). The S-States have different intrinsic PSII fluorescence yields (Gates et al., 2020; Zaharieva and Dau, 2019), so in an idealized sample, the four sequential S-States will be reflected by an ongoing periodic oscillation in ChlF. Recombination reactions, however, contribute to wasteful misses in the S-State cycling of individual PSII (Han et al., 2012). As misses occur, they progressively desynchronize S-State cycling across the PSII population, dampening the observed ChlF oscillation (de Wijn and van Gorkom, 2002) towards an averaged ChlF for the PSII population distributed across the S-States. An organism exhibiting synchronized S-State cycling sustained over more flash cycles indicates fewer photochemical misses, and thus higher conversion of sequentially spaced excitations to photosynthetic electron transport. By comparing the S-State cycling over flash cycles, of psychrophilic and temperate phytoplankton taxa, we sought to determine if psychrophilic diatoms and green algae show higher maintenance of progressive PSII function under extreme low light, thereby supporting stable extraction of electrons from water to support productive photosynthesis.

Materials and methods

Study Strains and Culturing Conditions

The nine study taxa, including polar and temperate strains of diatoms and green algae, and their respective culturing conditions, are summarized in Table 1. Previous research showed impacts of salinity on photosynthetic performance in some psychrophile strains (Kalra et al., 2020; Kalra et al., 2023; Stahl-Rommel et al., 2022), so we chose a range of polar and temperate taxa from both freshwater and saline environments. Further, since many polar *Chlamydomonas* species are found in both low temperature and high salinity,

we used a range of salinity growth conditions to address potentially confounding effects of salinity on our results.

Table 1: Study taxa and culture growth conditions.

Taxa	Line age	Origin	Growth °C	NaCl (mM)	NaCl (opt)	PAR (μ E)	Photoperio d (h)	Media
<i>Fragilariopsis cylindrus</i>	Diat om	Polar	0	500	500	10	24	F2
<i>Fragilariopsis cylindrus</i>	Diat om	Polar	0	500	500	1	24	F2
<i>Fragilariopsis cylindrus</i>	Diat om	Polar	6	500	500	10	24	F2
<i>Fragilariopsis cylindrus</i>	Diat om	Polar	6	500	500	1	24	F2
<i>Thalassiosira pseudonana</i>	Diat om	Temp erate	22	500	500	50	12	F2
<i>Chlamydomonas ICEMDV</i>	Gree n	Polar	4	70	500	10	24	BBM
<i>Chlamydomonas klinobasis</i>	Gree n	Polar	4	0	0	10	24	BBM
<i>Chlamydomonas klinobasis</i>	Gree n	Polar	4	10	0	10	24	BBM
<i>Chlamydomonas klinobasis</i>	Gree n	Polar	4	30	0	10	24	BBM
<i>Chlamydomonas malina</i>	Gree n	Polar	4	30	500	10	24	BBM
<i>Chlamydomonas malina</i>	Gree n	Polar	4	70	500	10	24	BBM
<i>Chlamydomonas malina</i>	Gree n	Polar	4	500	500	10	24	BBM
<i>Chlamydomonas priscui</i>	Gree n	Polar	4	30	500	10	24	BBM
<i>Chlamydomonas priscui</i>	Gree n	Polar	4	70	500	10	24	BBM
<i>Chlamydomonas priscui</i>	Gree n	Polar	4	500	500	10	24	BBM
<i>Chlamydomonas priscui</i>	Gree n	Polar	4	700	500	10	24	BBM
<i>Chlamydomonas euryale</i>	Gree n	Temp erate	24	30	500	10	24	BBM _NH 4
<i>Chlamydomonas euryale</i>	Gree n	Temp erate	24	70	500	10	24	BBM _NH 4

Taxa	Line age	Origin	Growth °C	NaCl (mM)	NaCl (opt)	PAR (μE)	Photoperiod (h)	Media
<i>Chlamydomonas euryale</i>	Green	Temperate	24	500	500	10	24	BBM_NH4
<i>Chlamydomonas reinhardtii</i>	Green	Temperate	24	0	0	10	24	BBM
<i>Chlamydomonas reinhardtii</i>	Green	Temperate	24	10	0	10	24	BBM
<i>Chlamydomonas reinhardtii</i>	Green	Temperate	24	30	0	10	24	BBM
<i>Chlorella vulgaris</i>	Green	Temperate	22	500	500	50	12	F2
<i>Chlorella vulgaris</i>	Green	Temperate	22	500	500	50	12	BG11

Fragilariopsis cylindrus is a psychrophilic (Fiala and Oriol, 1990) pennate diatom measuring 15-55 μm which forms large blooms in the high salinity and subzero temperatures of Arctic and Antarctic sea-ice systems (Otte et al., 2023; Cefarelli et al., 2010), and is a keystone species for polar ecosystems (von Quillfeldt, 2004; Otte et al., 2023; Kang and Fryxell, 1992). Sequenced in 2017, *Fragilariopsis cylindrus* has since become the model psychrophilic diatom species for investigating molecular and physiological mechanisms of cold adaptation (Mock et al., 2017). Conversely, *Thalassiosira pseudonana* (*Cyclotella nana*) is a centric diatom (2.5-15 μm) found in diverse freshwater, coastal, brackish, and marine habitats (Poulsen and Kröger, 2023), across a range of salinities and temperatures (4–25°C) (Poulsen and Kröger, 2023). *Chlamydomonas* sp. ICE-MDV and *Chlamydomonas priscui* are halotolerant algae isolated from the perennially ice-covered hypersaline Lake Bonney, in McMurdo Dry Valleys, Antarctica (Cook et al., 2019; Stahl-Rommel et al., 2022). With 15 to 20 μm biflagellate cells, *Chlamydomonas* sp. ICE-MDV dominates the shallow photic zone, where it experiences relatively higher irradiance,

extreme nutrient limitation, and lower salinity (Cook et al., 2019; Li et al., 2016). The smaller *Chlamydomonas priscui* dominates the deep photic zone, characterized by permanent low temperatures, low irradiance, and high salinity (Cvetkovska et al., 2022; Hüner et al., 2023). *Chlamydomonas malina* is a marine microalga isolated from the Arctic Beaufort Sea, around 10 µm in length and 5 µm in width, growing optimally at 4°C (Balzano et al., 2012; Morales-Sánchez et al., 2020). *Chlamydomonas klinobasis* strain 041-99 is a freshwater microalga isolated from an ephemeral snowfield on Svalbard Island, Norway, of around 12 x 6 µm (P. Osmer, unpub), growing from 2-6°C (Hulatt et al., 2017). The temperate *Chlamydomonas reinhardtii* is a model green alga approximately 10 µm in diameter, found in freshwater soil and aquatic environments with an optimal temperature range of 20-32°C (Sasso et al., 2018; Xie et al., 2013). The temperate marine *Chlamydomonas euryale* is approximately 4 µm in diameter, isolated from various locations within the northern Atlantic and Pacific, including the Yellow Sea, with a known growth temperature range of 18-26°C Carrasco Flores et al. (2021). *Chlorella vulgaris*, ranging from 2 µm to 10 µm in size, is primarily found in freshwater environments and grows optimally at 27°C (Wiel et al., 2017; Leyva et al., 2014).

Thalassiosira pseudonana and *Chlorella vulgaris* were cultured by Naaman Omar (Mount Allison University); *Chlamydomonas* strains were cultured by Mackenzie Poirier (Cvetkovska Lab, University of Ottawa) as reported previously (Poirier et al., 2025); while *Fragilariopsis cylindrus* was cultured by Sébastien Guérin (Takuvik International Research Laboratory, Université Laval).

Single Turnover Variable Chlorophyll Fluorescence

We used series of single turnover, saturating, flashes to induce variable chlorophyll fluorescence (ChlF) (Kolber, 1998), and evaluate progressive desynchronization of the S-State cycle in several polar and temperate phytoplankton species acclimated to a range of growth temperatures. We loaded a 3 mL sample of culture into a temperature-controlled water-jacketed quartz cuvette (Cafferty's Scientific Glassblowing, Anola, Manitoba, Canada), placed within the measurement chamber of a Soliense fluorometer (Version LIFT-REM 1.0, Soliense Inc., Shoreham NY, USA). The apparatus was covered to block out incident light, for a minimum of 30 seconds prior to measurements. In the dark PSII reaction centres relax to ground state, and are open for photochemistry upon receipt of an absorbed photon, with a low fluorescence yield (Schuback et al., 2021). In parallel the OEC relaxes primarily to S1 (Han et al., 2008), so the S-States of the population of PSII largely synchronize. The sample was then exposed to a series of 20-32 short, high irradiance evenly spaced flashes centred at 445 nm. Excitation at 445 nm preferentially excites chlorophyll a, thereby limiting the influence of different light harvesting complexes across taxa. Each flash comprised a rapid series of 50-70 sub-saturating flashlets of 1.6 μs and $\sim 30000 \mu\text{mol photons m}^{-2} \text{s}^{-1}$, delivered every 4.1 μs , over a total 205 - 287 μs per flash (Schuback et al., 2021). These flashlets cumulatively deliver a photon to each PSII in the population, with each PSII ideally undergoing a single round of photochemistry and passing an electron downstream to Q_A , closing PSII for photochemistry for $\sim 1000 \mu\text{s}$, the lifetime (τ) for re-opening by downstream electron transport (Dau and Haumann, 2007; Schuback et al., 2021). Closing the photochemistry pathway transiently redirects a greater proportion of additional incoming excitation to ChlF, driving ChlF progressively from minimum (F_0) towards a

maximum (F_M) as the PSII centres close (Schuback et al., 2021). For each flash-induced fluorescence rise, F_0 and F_M were extracted using a fitting model (LIFT software version 22.11.11, Solisense Inc) (Kolber, 1998). F_0 and F_M were then used to derive the maximum quantum yield of photochemistry for open PSII (Schuback et al., 2021), a secondary ChlF parameter calculated as:

$$\frac{F_V}{F_M} = \frac{F_M - F_0}{F_M} \quad (1)$$

For each culture sample we adjusted the number and excitation power of flashlets per flash to achieve closure of ~ 0.95 of PSII complexes, as estimated by the fitting of the chlorophyll fluorescence induction (Kolber, 1998; Schuback et al., 2021). Ideally the saturating flash would achieve complete closure of the PSII population, but concomitant re-opening of PSII centres by downstream electron transport results in a steady state ratio of mainly closed PSII to a fraction of open PSII, even at F_M . The flash power required to drive the PSII population to closure depends largely upon the effective absorption cross-section of photosystem II (σ_{PSII}), itself estimated from the ChlF induction curve during the saturation phase (Kolber, 1998). σ_{PSII} represents the probability of light capture by the PSII antenna bed associated with the dark-adapted PSII (Schuback et al., 2021; Xu et al., 2018). When σ_{PSII} for a sample is large, the higher probability of excitation delivery to PSII means fewer, or weaker, flashlets are required to sequentially saturate the population of PSII, while limiting excitation stress or double turnovers of PSII associated with excess light (Xu et al., 2018). Conversely, a sample with a smaller σ_{PSII} requires more numerous, or brighter, flashlets, to drive the proportion of closed reaction centres progressively towards

saturation. Because we used excitation at 445 nm, delivered through PSII-associated chlorophyll a, our σ_{PSII} were quite consistent across taxa (Xu et al., 2018), even though the light harvesting complexes differ between the green and diatom taxa. If the brightness and number of sub-saturating flashlets are appropriate for the culture, the fluorescence will reach a plateau after approximately 70% of the flashlets are applied (Kolber, 1998).

We kept the single turnover excitation flash train as short as feasible, to avoid excessive excitation, and to limit the probability of ‘double hits’ (Han et al., 2012, 2022); re-openings of PSII, followed by re-closures of PSII complexes by a second round of photochemistry, which contribute to desynchronization of the progression of the PSII complexes through S-States. Furthermore we did not use probe flashlets to track re-opening of PSII after closure by each saturating flash (Kolber, 1998), to limit differential re-openings and re-excitations across the PSII population. We thereby forewent analyses of PSII re-opening kinetics.

As sequential flashes are applied to the culture, each individual PSII is ideally driven through the four S-States (Fig 1) (Dau and Haumann, 2007). As the OEC of PSII moves between S-States, it alters the system kinetics and free energy (Vinyard et al., 2013), so the yield of ChlF varies between S-States (Gates et al., 2020), and we can use ChlF to monitor S-State progression, until the PSII population desynchronizes to a distribution of S-States with an averaged ChlF.

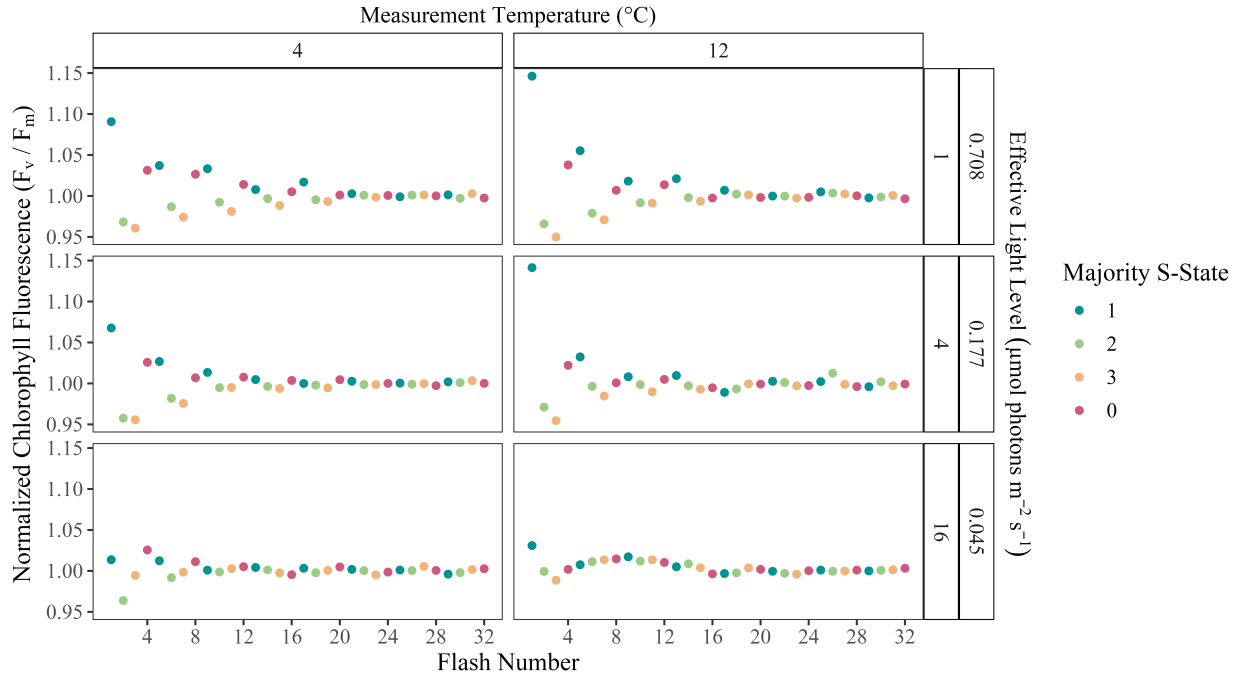


Fig 1: Representative oscillations of the maximum quantum yield of PSII photochemistry over a Series of Single Turnover Saturating Flashes applied to *Chlamydomonas priscui*. F_v/F_m was normalized to the average value over the series, measured at 4 or 12 °C, with spacing of 1, 4, or 16 s between sequential flashes, excitation rates equivalent to irradiance with 0.708, 0.177 or 0.044 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$. Symbol colour indicates inferred majority S-State, based upon relaxation to S1 during the dark incubation preceding the flash sequence.

Measurements

By evaluating the oscillations in ChlF (Fig 1), we tracked S-State cycling of polar and temperate taxa of diatoms and green algae under a range of measurement temperatures and effective light levels, after growth under a range of salinities (Table 4), to determine if polar taxa maintain synchronized PSII function, as an index of their capacity for electron transport under low light, by suppressing wasteful photochemical misses. Measurement temperatures ranged from 0 to 28°C, depending on the taxa (Table 4).

Increasing the spacing between sequential flashes decreases the repeat rate for excitations arriving at PSII, thereby approximating a decrease in excitations delivered to

PSII per second under a decrease in continuous irradiance (Fig 1). Our instrument control limited our minimum flash spacing to 1 s, so cultures were evaluated at flash spacings of 1, 2, 4, 8, and 16 seconds (Table 4). The equivalent effective light levels were then estimated using the σ_{PSII} determined for each culture as follows:

$$\text{Light}(\text{photons m}^{-2} \text{ s}^{-1}) = \frac{1}{\text{flash spacing (s)}} \times \frac{1}{\sigma_{PSII} (\text{m}^2 \text{ photons}^{-1})} \quad (2)$$

$1/\sigma_{PSII}$ is multiplied by flash spacing (s) to convert the probability that an individual incident photon is absorbed by PSII, into an equivalent photon flux density, i.e. to estimate effective equivalent light availability serving PSII photochemistry.

Since the population of PSII starts predominantly from S1 in the dark acclimated state, the population of PSII will initially cycle synchronously through S States, reflected by an oscillation in chlorophyll fluorescence with a period of four (Fig 1)(de Wijn and van Gorkom, 2002). Photochemical misses, including wasteful recombination reactions causing loss of a charge separation in a PSII, will cause a missed step in the S-State cycling of an individual PSII. As more photochemical misses occur, desynchronization of S-State cycling among the population of PSII will scramble the periodic changes in ChlF, dampening the observed oscillation of the population-level fluorescence (de Wijn and van Gorkom, 2002; Han et al., 2022). Prolonged four-step synchronous cycling of PSII fluorescence thus reflects a lower probability of desynchronization at each excitation, and thus a lower probability of losses through photochemical misses. Photochemical double hits also contribute to

desynchronization (Han et al., 2022; Han et al., 2008), but should not vary strongly across changes in flash spacing nor measurement temperature within a sample.

Analytical Methods

Data was processed using R version 4.3.2 (R Core Team, 2023) and RStudio version 2023.12.0+369 (Posit team, 2025) on the x86_64-apple-darwin20 (64-bit) platform and running under macOS Sonoma 14.3.1. Fluorescence data files generated by LIFT software were imported, tidied and combined with metadata on each culture using the tidyverse (Wickham, 2023), lubridate (Spinu et al., 2024), and googlesheets4 (Bryan, 2023) packages. The doBy (Halekoh and Højsgaard, 2025), and WaveletComp (Roesch and Schmidbauer, 2018) packages were used for wavelet analyses. The mgcv (Wood, 2011) and mgcViz (Fasiolo et al., 2020) packages were used for generalized additive modelling, while the ggplot2 (Wickham, 2016), ggh4x (van den Brand, 2025), and metR (Campitelli, 2021) packages were used for data visualization. Statistical results tables were done using the packages 'broom' (Robinson et al., 2025), 'knitr' (Xie, 2025), and 'kableExtra' (Zhu, 2024).

Wavelet Transformations

The fluorescence data yields a time series of F_V/F_M over 32 sequentially applied flashes, for each flash spacing, and each measurement temperature, for each culture sample. The F_V/F_M time series were analyzed for each combination of strain, growth conditions, measurement temperature, and flash spacing, using wavelet transformations (Theis and Meyer-Bäse, 2010), as exemplified in (Fig 2). Unlike traditional methods, wavelet analysis does not assume that the statistical properties of a time series are constant. Instead, wavelet transformations locally decompose the signal across different

time scales and estimate spectral characteristics as a function of time (Cazelles et al., 2008). By examining the frequency and wavelet power spectra, we can uncover the dominant patterns in the data (Theis and Meyer-Bäse, 2010).

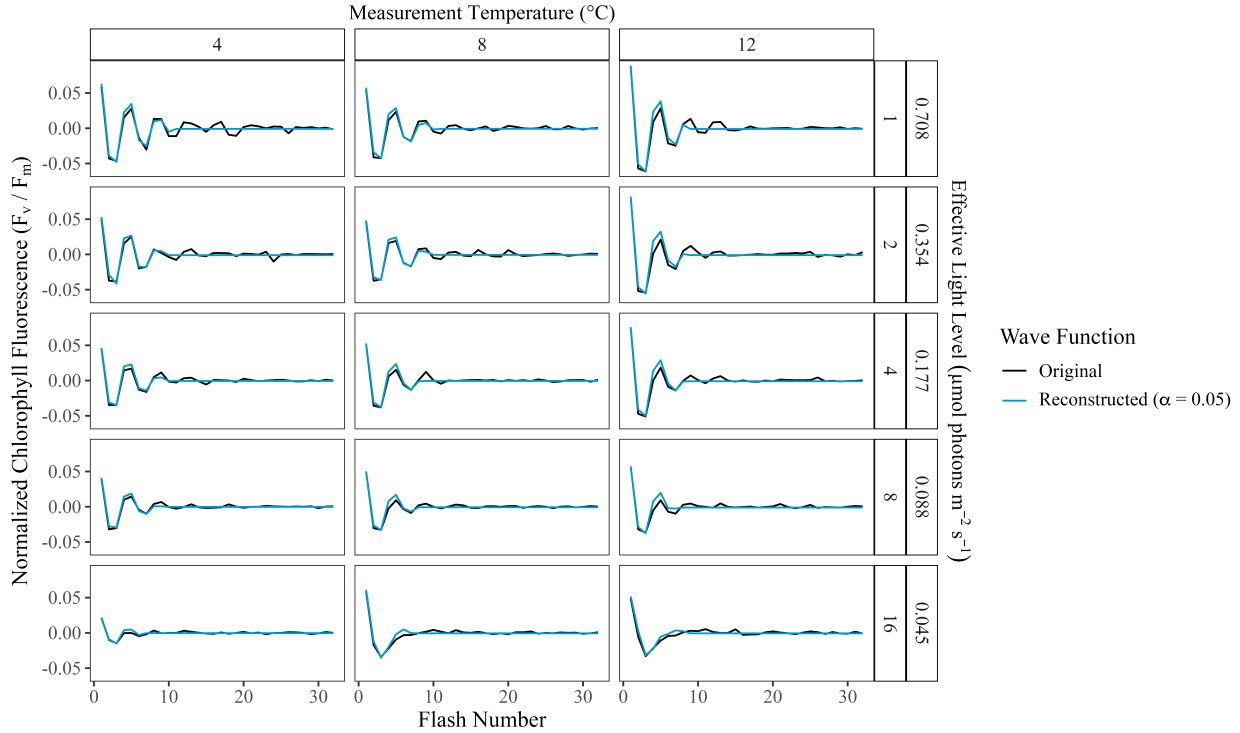


Fig 2: Representative wavelet transformations of Antarctic *Chlamydomonas priscui* variable chlorophyll fluorescence (F_V/F_M) measured at 4, 8 or 12°C, over 32 consecutive single turnover saturating flashes, applied at spacings of 1, 2, 4, 8, 16 s, equivalent to photon delivery rates to PSII achieved under effective light levels from 0.708 down to 0.045 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$.

The wavelet transformation involves computing the wavelet power spectrum of the standardized time series using the Morlet wavelet (Theis and Meyer-Bäse, 2010). The statistical significance of the periodic components in the time series was then calculated using a simulation algorithm. Surrogate time series are generated based on a white noise model, consisting of uncorrelated random values with constant mean and variance. The wavelet transform of the data is compared with the white noise model, to estimate p-values

for whether the observed periodic components are statistically significant (Theis and Meyer-Bäse, 2010). Statistical significance of the wavelet power at a periodicity of four indicates the culture is exhibiting periodic oscillations in chlorophyll fluorescence, reflecting synchronous S-State cycling across the PSII population. For wavelets exhibiting detectable S-State cycling, we generated a reconstruction, limited to the sub-set of flashes showing a signal that was statistically different from the average value of F_V/F_M . We then extracted a Damping Index as the number of flashes applied before desynchronization results in the reconstructed wavelet dropping below a detection threshold significance of $p=0.05$ above random noise. This Damping Index indicates how many successive photochemical charge separations occur in the PSII population before recombinations, other photochemical misses, or photochemical double hits, desynchronize the S-State cycle to a randomized distribution of S-States across the population.

Generalized Additive Modelling

The patterns in the persistence of S-State cycling within taxa were then modeled using the nonparametric method of generalized additive modelling (GAM). GAMs fit a model to predict the damping index based on a tensor product smooth of the two predictors, measurement temperature, and equivalent effective light level. The response variable is linked to the independent variables using a smoothing function, where many localized polynomials are joined to form a piecewise function called a spline (Pinilla and Negrín, 2021). For each strain, GAM models were fit to the data using the restricted maximum likelihood method (REML).

The GAM model examines the response of S-State damping to the measurement temperature (°C) and the equivalent effective light level ($\mu\text{mol photons m}^{-2} \text{ s}^{-1}$) for the measurement flash spacing. The generalized additive models were fitted using the R-package *mgcv* (Wood, 2025). Only S-State damping estimates for which the amplitude of standard error was smaller than 320% of the fitted S-State damping were included in the GAM. Models were validated by verifying the choice of basis dimensions (k) and evaluating the residual plots (Schoenig et al., 2023) using the function `gam.check` from the R-package *mgcv* (Wood, 2025) (Table ??). Based on the fitted models, the damping index can then be predicted for other combinations of measurement temperature and effective light (Wood, 2017), represented with a contour plot.

Analyses of Variance

Analyses of Variance (ANOVA) were run using the `aov` function (R Core Team, 2023) to detect effects of taxa, measurement effective light levels (derived from flash spacing and σ_{PSII}), measurement temperature, and growth salinity, upon the Damping Index.

Results

Single Turnover Variable Chlorophyll Fluorescence

Exposing phytoplankton cultures to a series of 20-32 successive single turnover, saturating flashes produced oscillations in the maximum quantum yield of photochemistry of PSII, as estimated through the secondary chlorophyll fluorescence parameter F_V/F_M , normalized to 1 for each run by dividing by the averaged value for each measurement run, after oscillations damped out (Table 4). Initially, the majority of the dark-adapted

population of PSII is at S1, with a smaller fraction at S0 (Gates et al., 2020; de Wijn and van Gorkom, 2002). As shown in representative data from polar *Chlamydomonas priscui* (Figures 1, 2), grown at optimal salinity, the time series of F_V/F_M over successive flashes reveals consistent variations in fluorescence yield as the predominant S-States, with different yields for ChlF, follow each other across the PSII within the population. The amplitudes of the ChlF oscillations, however, decline progressively over successive flashes. The persistence of oscillations decreases with wider spacing between sequential flashes, equivalent to decreasing effective light levels, and at higher measurement temperatures. The full set of S-State time courses captured across taxa and growth salinities are in Supplemental Data.

The conversions of flash spacing to equivalent effective light levels gave similar ranges of effective light levels applied to each strain (Table 4). For comparison, full sunlight at the sea surface is $\sim 2000 \mu\text{mol photons m}^{-2}\text{s}^{-1}$, while the threshold for the bottom of the photic zone supporting photosynthetic productivity in the oceans is conventionally defined as $\sim 20 \mu\text{mol photons m}^{-2}\text{s}^{-1}$ (Raven et al., 2000). Therefore, our measurement light ranges are ~ 5 orders of magnitude lower than full sunlight, and ~ 2 -3 orders of magnitude below the traditional minimum photic zone threshold, but nevertheless span light ranges found to support phytoplankton productivity in field studies (Hancke et al., 2018; Randelhoff et al., 2020; Hoppe et al., 2024).

Wavelet Analysis

Wavelet transformations were computed for the fluorescence time series of each unique combination of measurement temperature, flash spacing (and thus equivalent

effective light), growth temperature, and species or strain. Assessing the wavelet power of a 4-step periodicity across conditions, key trends emerge. As exemplified by the Antarctic green alga *Chlamydomonas priscui* (Figure 3), the average wavelet power declines with increasing flash spacings, equivalent to decreasing effective light levels, and also declines with increasing measurement temperatures.

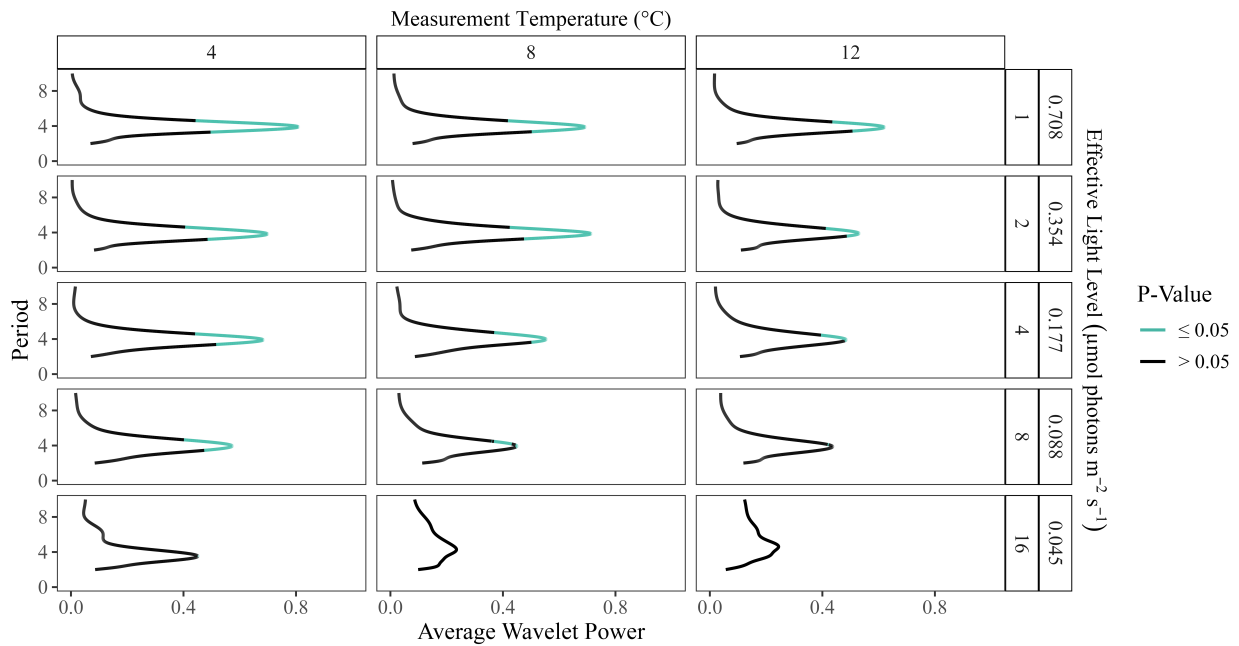


Fig 3: Sample plot of wavelet powers by period of oscillations in the maximum quantum yield of photochemistry Antarctic green algae *Chlamydomonas priscui*, grown at 4°C and 500 mM NaCl, and measured across a range of measurement temperatures and flash spacings, with the equivalent effective light levels. Regions where wavelet power reached statistical significance ($p < 0.05$) are shaded in blue.

In contrast, in the temperate green algae *Chlamydomonas reinhardtii* (Figure 4) both grown, and measured, at warmer temperatures, the wavelet power is consistently lower, showing a weaker 4-step periodicity of ChlF in the temperate taxa, which only reaches statistical significance at shorter flash spacings, with higher equivalent effective light levels.

Note that at a common measurement temperature of 12°C *Chlamydomonas reinhardtii* and *Chlamydomonas priscuii* show similar wavelet power patterns.

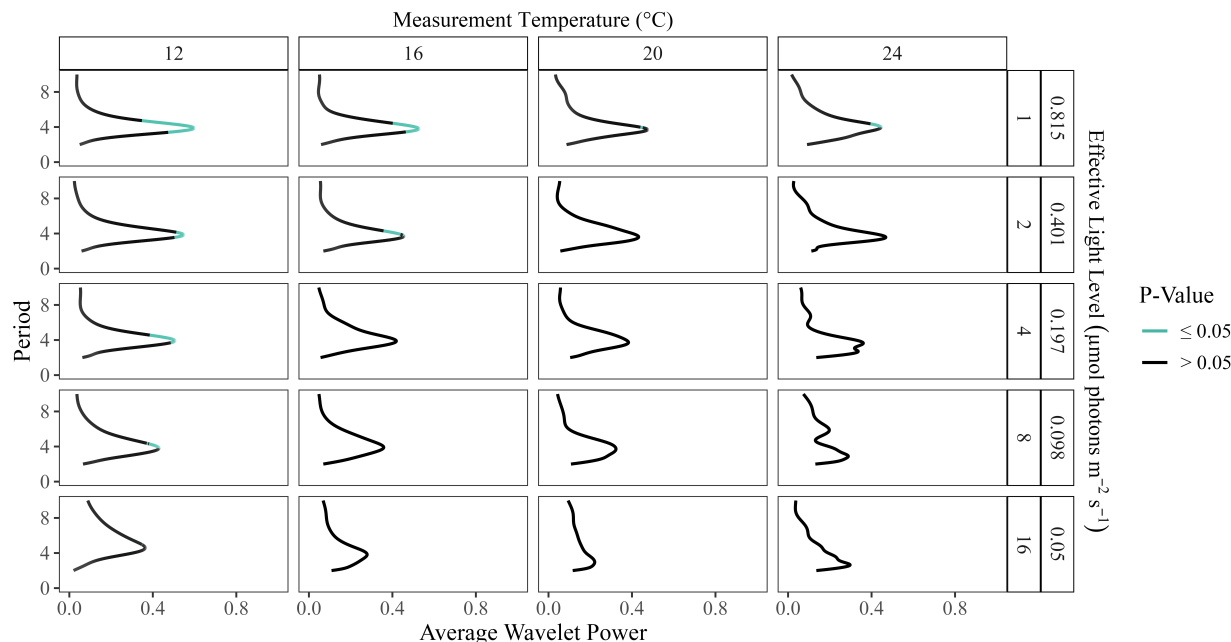


Fig 4: Sample plot of wavelet powers by period of oscillations in the maximum quantum yield of photochemistry in the temperate green algae *Chlamydomonas reinhardtii*, grown at 24°C and 0.43 mM NaCl, and measured across a range of measurement temperatures and flash spacings, with the equivalent effective light levels. Regions where wavelet power reached statistical significance ($p < 0.05$) shaded in blue.

Generalized Additive Modelling of Damping of 4-step oscillations By Measurement Temperature

Predictions from generalized additive modelling were generated for the damping of S-State-induced chlorophyll fluorescence oscillations, as predicted by the tensor product smooth of the measurement temperature (°C) and the equivalent effective light level ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$) set by flash spacing, for each strain; Table 4; Table 5. All of the model fits explained over 50% of the variation in the response variable, damping index, except the GAM fits for *C. euryale*, which failed to fit. Both diatom taxa exhibited the longest predicted periodic oscillations in ChlF at higher effective light levels and lower measurement

temperatures. Notably, the polar *Fragilariopsis cylindrus* sustained cycling longer than its temperate counterpart, *Thalassiosira pseudonana*, under comparable conditions. This disparity was particularly prevalent at measurements taken above the growth temperature and when longer spacing between flashes produced lower effective light levels (Fig 5), where *Thalassiosira pseudonana* cultures did not retain the significant 4-step oscillation in ChlF indicative of synchronized S-State cycling.

Similarly, the GAM outputs varied among the green algae tested (Table 5). The polar, saline strains *Chlamydomonas* ICE-MDV, *Chlamydomonas malina* and *Chlamydomonas priscui* showed GAM model predictions of maximum damping indices of ~ 10-11 flashes, centred at lower measurement temperatures but extending over a wide range of effective light levels (Fig 6). The polar freshwater *Chlamydomonas klinobasis* also showed maximum damping indices at lower measurement temperatures, across a range of effective light levels. Further, much like the temperate diatoms, the freshwater temperate algae *Chlamydomonas reinhardtii* and *Chlorella vulgaris* did not exhibit significant periodic oscillations in ChlF at measurement temperatures near or above their growth temperature, under longer flash spacings equivalent to lower light. The marine temperate *Chlamydomonas euryale* did not generate a statistically significant GAM model, consistent with limited evidence for periodic oscillations from *Chlamydomonas euryale*, which, also showed $F_V F_M$ somewhat lower than other *Chlamydomonas* (Table 4).

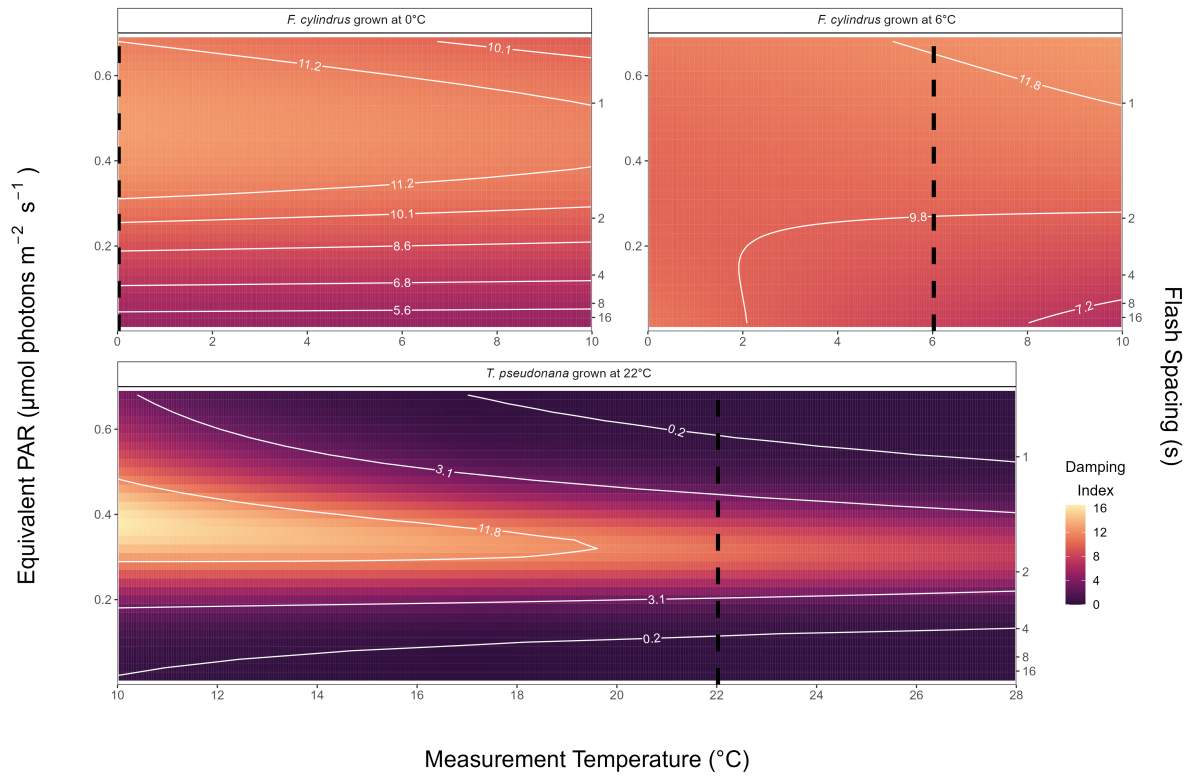


Fig 5: GAM models for polar and temperate diatoms, of consecutive flashes before damping of SState induced chlorophyll fluorescence oscillations. GAM model predicted by the measurement temperature ($^{\circ}\text{C}$) and the effective light level ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$, estimated from flash spacings in seconds and σ_{PSII}). Dashed vertical lines represent the growth temperatures. Measurement temperatures varied across taxa but the temperature axis scalings are comparable across plots.

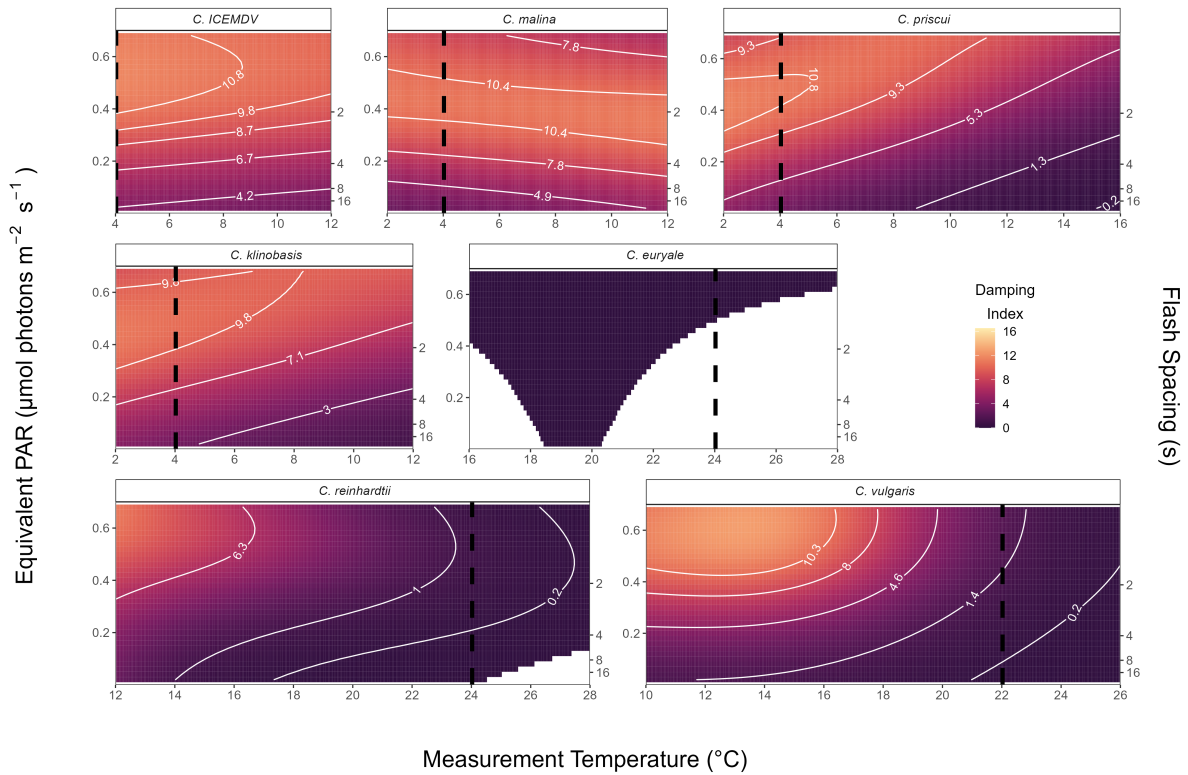


Fig 6: GAM models for polar and temperate green algae, of consecutive flashes before damping of SState induced chlorophyll fluorescence oscillations. GAM model predicted by the measurement temperature ($\Delta^{\circ}\text{C}$) and the effective light level ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$, estimated from flash spacings in seconds and σ_{PSII}). Dashed vertical lines represent the growth temperatures. Measurement temperatures varied across taxa but the temperature axis scalings are comparable across plots.

The *Chlamydomonas* strains we used can survive under a range of salinities. Furthermore, across several species of *Chlamydomonas* growth under higher salinities increases photosynthetic resilience during stress (Kalra et al., 2020; Kalra et al., 2023; Stahl-Rommel et al., 2022). Therefore, to test for effects of salinity acclimation upon the performance of PSII under extreme low light, we grew each *Chlamydomonas* strain across three growth-permissive salinities (Table 1) and then performed saturating flash trains over a range of flash spacings, and at measurement temperatures flanking the growth temperature for each taxa. We ran an ANOVA for damping indices testing effects of taxa,

effective equivalent light level (derived from the saturating flash spacing and σ_{PSII}), measurement temperature, and the growth concentration of NaCl, with interactions among all factors (Table ??). As expected, taxa, effective equivalent light level and measurement temperature all showed strongly significant influences on damping indices. NaCl concentration also showed a significant influence on damping indices across strains, with significant interaction between NaCl and effective equivalent light level.

Table 2: Analyses of Variation of damping indices testing effects of *Chlamydomonas* taxa, effective equivalent light level ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$, estimated from flash spacings in seconds and σ_{PSII}), measurement temperature, and the growth concentration of NaCl, with interactions among all factors.

term	df	sumsq	meansq	statistic	p.value
Strain	4	2128	532	78.5	0.00
LightLevel_u e	1	577	577	85.1	0.00
Temp_C	1	488	488	72.0	0.00
NaCl_mM	1	35	35	5.2	0.02
Strain:LightL evel_ue	4	369	92	13.6	0.00
Strain:Temp_ C	4	203	51	7.5	0.00
LightLevel_u e:NaCl_mM	1	27	27	4.0	0.05
Strain:LightL evel_ue:Temp_ _C	4	74	18	2.7	0.03

We then ran the ANOVA for each taxa separately (Table 3). Within a taxa only *Chlamydomonas priscui* showed a significant effect of NaCl on damping indices, suggesting the statistical influence of NaCl detected across taxa derives from different taxa growing at different salinities and showing different responses of damping indices, rather than a direct effect of NaCl on damping indices in most strains.

Table 3: Analyses of Variation for each *Chlamydomonas* species testing effects upon damping indices of effective equivalent light level ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$, estimated from flash spacings in seconds and σ_{PSII}), measurement temperature, and the growth concentration of NaCl, with interactions among all factors.

Strain	term	df	sums q	means q	statistic	p.value
<i>Chlamydomonas_euryale</i>	LightLevel_ue	1	30	30	5.1	0.03
<i>Chlamydomonas_klinobasis</i>	LightLevel_ue	1	550	550	77.5	0.00
<i>Chlamydomonas_klinobasis</i>	Temp_C	1	137	137	19.2	0.00
<i>Chlamydomonas_malina</i>	LightLevel_ue	1	140	140	14.5	0.00
<i>Chlamydomonas_priscui</i>	LightLevel_ue	1	255	255	31.3	0.00
<i>Chlamydomonas_priscui</i>	Temp_C	1	476	476	58.5	0.00
<i>Chlamydomonas_priscui</i>	NaCl_mM	1	40	40	4.9	0.03
<i>Chlamydomonas_reinhardtii</i>	Temp_C	1	18	18	5.2	0.03

Discussion

Across all study strains ChlF oscillations, provoked by a series of single turnover saturating flashes, showed wavelet power at a period of four, that declined with increasing measurement temperatures and increasing spacing of flashes, equivalent to decreasing equivalent effective light. Thus, sustained synchronized S-State cycling of PSII decayed faster under higher measurement temperatures and lower equivalent effective light (Figure 3) (de Wijn and van Gorkom, 2002). While light absorption and excitation energy transfer within PSII are not temperature-sensitive, the redox reactions associated with downstream electron transport are highly temperature-dependent (Hüner et al., 2023), as are probabilities of charge recombinations (Ivanov et al., 2006; Han et al., 2022). Temperature dependences of recombinations may interact with acclimatory mechanisms to regulate photosynthetic electron flow, S-State transitions, and energy partitioning (Hüner et al., 2023). In future, treatments with plastoquinone-analog inhibitors of PSII such as

dichlorophenyl-dimethylurea (DCMU) might help dissect distinctions among responses across psychrophile and temperate strains strains.

The desynchronization of S-State cycling among the PSII in a population indicates that sufficient photochemical misses, including charge recombinations, have taken place to create a PSII population with a randomized distribution of S-States. Thus, if this desynchronization occurs after fewer consecutive flashes, it signifies an increased proportion of PSII undergoing photochemical misses for each flash. By inference, PSII populations with increased incidence of energetically wasteful misses, including charge recombinations, such as those under high temperatures and low effective light levels, are less efficient in their photosynthetic energy conversion (Rappaport et al., 2005), which becomes a critical factor for maintaining photosynthesis under low light, with wide spacings between sequential excitations of each PSII.

These results are consistent with previous literature evaluating the responses of photochemical misses to light conditions. As light levels decline, there are longer intervals between successive PSII excitations, pushing fewer electrons through the electron transport chain (Keren et al., 1997). Consequently, the probability of energetically wasteful charge recombinations is higher over the longer intervals between excitations, leading to weaker maintenance of S-State cycling (Keren et al., 1997; de Wijn and van Gorkom, 2002). These findings are consistent with recombination reactions decreasing as measurement temperatures (Ivanov et al., 2006) fall below the activation temperatures of the recombinations. In contrast, in PSII-enriched membrane fractions isolated from spinach,

the average miss probability of S-State transitions was highest at -10 °C and lowest at 10 °C (Han et al., 2022).

Chlamydomonas sp. ICE-MDV and *Chlamydomonas priscui* originate from Lake Bonney in Antarctica, but previous work found differential responses to light among the strains. Unlike the obligately shade-adapted *Chlamydomonas priscui* that is only present in the deep photic zone, *Chlamydomonas* sp. ICE-MDV has been isolated from various depths within the lake (Li and Morgan-Kiss, 2019) including shallow and seasonally open waters (Sherwell et al., 2022) where it encounters a wider range of light conditions. Thus, in contrast with *Chlamydomonas priscui* that has a “locked” physiology, *Chlamydomonas* sp. ICE-MDV has retained the ability to balance excitation pressure through state transitions (Kalra et al., 2023), can modify its photosynthetic apparatus due to nutrient availability (Cook et al., 2019), exhibits phototactic motility in response to light signals (Poirier et al., 2023) and can successfully acclimate to a range of wavelengths and light levels (Poirier et al., 2025). It is evident that physiological differences exist among polar species, likely as a result of life in comparably cold, but otherwise differing environments.

Nevertheless, strong, sustained periodicity of ChlF emissions is a common trait across the diverse polar strains of diatoms and green algae (Figures 3; 4; 5; 6. Further, the polar taxa exhibited significant 4-step ChlF oscillations under a broader range of measurement conditions around their growth temperatures than their temperate counterparts. These findings illustrate that polar phytoplankton species, measured at and around growth temperatures typical for their native environments, show higher capacity to maintain synchronized S-State cycling than do temperate strains around their optimal

growth temperatures. The rates of photochemical misses, notably through charge recombinations, relative to productive photochemistry, increase under lower light and higher measurement temperatures (Keren et al., 1997; Ivanov et al., 2006). Thus, sustained synchronized S-State cycling reflects suppression of energetically wasteful photochemical misses in the polar strains, as measured at the low growth temperatures possible for polar strains. Stable S-State cycling with minimal energy losses through charge recombination or other forms of electron cycling within PSII (Ananyev et al., 2017; Zournas et al., 2023) can support linear electron flow, sustaining ATP and NADPH production, and minimizing the risk of photodamage to the photosynthetic machinery (Rappaport et al., 2005; Keren et al., 1997). Thus, sustaining PSII function even under widely spaced excitations appears as an emergent property supporting the productivity of polar phytoplankton under the ice during the polar night.

Salinities vary widely across the habitats of the tested strains and fluctuate within some habitats during ice formation and melting (Morgan-Kiss et al., 2016; Smoot and Hopcroft, 2017; Carton et al., 2015). Higher salinity may have photoprotective properties in the polar strains of *Chlamydomonas* (Kalra et al., 2020; Kalra et al., 2023; Stahl-Rommel et al., 2022). NaCl concentration showed an co-variate influence on dampening indices when analyzed across all tested *Chlamydomonas* taxa, but when tested within taxa, NaCl only showed an influence within *Chlamydomonas priscui*. Out of all the *Chlamydomonas* species used here, *Chlamydomonas priscui* exhibits the widest range of permissive growth salinities (10 mM - 1.2 M) and salinity significantly affects photosystem partitioning and thylakoid organization in *Chlamydomonas priscui* (Szyszka-Mroz et al., 2019). It is important to note that faster desynchronization of S-State cycling does not indicate a loss nor instability of

PSII function *per se*. Desynchronization rather reflects randomization of the S-States across the PSII population. Prolonged synchronization under low effective light levels is thus an indicator of the capacity of PSII to exploit widely spaced photons under low light, not the stability of PSII function itself.

Understanding and detecting mechanisms enabling polar phytoplankton to sustain slow, but significant, productivity under the ice in the winter can help predict the changing dynamics of spring phytoplankton blooms, in the face of rapid warming (Ardyna and Arrigo, 2020). Beyond direct temperature changes, polar aquatic ecosystems are experiencing declines in sea ice extent and thickness, escalating freshwater inputs, acidification, and increased winds and storms (Ardyna and Arrigo, 2020; Cvetkovska et al., 2022). These pressures are altering the productivity and seasonal peaks of phytoplankton blooms (Ardyna and Arrigo, 2020). Maintaining photosynthetic energy conversion over winter is crucial for the timing and speed of spring bloom initiation (Hancke et al., 2018). Our findings indicate that low growth temperature is a factor enabling phytoplankton to exploit extreme low light. Taxa with this capacity then possess a competitive advantage in quickly initiating spring growth, giving them first access to the nutrients required to form an extensive bloom and in turn, exert bottom-up effects on changing polar ecosystems (Ardyna and Arrigo, 2020).

Acknowledgements

Supporting information

Table 4: Single turnover saturating flash measurement conditions, with key growth conditions, equivalent effective light levels ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$), mean σ_{PSII} (A^2) and mean F_vF_m for each measurement

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _euryale	1	16	0.71	24	30	238	0.47
Chlamydomonas _euryale	1	16	0.74	24	70	224	0.48
Chlamydomonas _euryale	1	16	0.76	24	500	222	0.44
Chlamydomonas _euryale	1	20	0.70	24	30	242	0.45
Chlamydomonas _euryale	1	20	0.71	24	70	239	0.46
Chlamydomonas _euryale	1	20	0.75	24	500	231	0.45
Chlamydomonas _euryale	1	24	0.72	24	30	232	0.43
Chlamydomonas _euryale	1	24	0.75	24	70	223	0.62
Chlamydomonas _euryale	1	24	0.76	24	500	224	0.55
Chlamydomonas _euryale	1	28	0.69	24	30	248	0.43
Chlamydomonas _euryale	1	28	0.71	24	70	238	0.44
Chlamydomonas _euryale	1	28	0.72	24	500	234	0.42
Chlamydomonas _euryale	2	16	0.35	24	30	234	0.46
Chlamydomonas _euryale	2	16	0.37	24	500	224	0.44
Chlamydomonas _euryale	2	20	0.34	24	30	241	0.45
Chlamydomonas _euryale	2	20	0.35	24	70	241	0.46
Chlamydomonas _euryale	2	20	0.36	24	500	234	0.45
Chlamydomonas _euryale	2	24	0.37	24	70	228	0.62

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _euryale	2	24	0.37	24	500	225	0.54
Chlamydomonas _euryale	2	24	0.37	24	30	227	0.42
Chlamydomonas _euryale	2	28	0.32	24	30	262	0.43
Chlamydomonas _euryale	2	28	0.35	24	70	239	0.43
Chlamydomonas _euryale	2	28	0.35	24	500	236	0.42
Chlamydomonas _euryale	4	16	0.17	24	30	245	0.47
Chlamydomonas _euryale	4	16	0.18	24	70	232	0.48
Chlamydomonas _euryale	4	16	0.18	24	500	229	0.46
Chlamydomonas _euryale	4	20	0.17	24	30	248	0.45
Chlamydomonas _euryale	4	20	0.17	24	70	246	0.46
Chlamydomonas _euryale	4	20	0.18	24	500	235	0.45
Chlamydomonas _euryale	4	24	0.18	24	30	235	0.43
Chlamydomonas _euryale	4	24	0.18	24	70	226	0.63
Chlamydomonas _euryale	4	24	0.18	24	500	232	0.54
Chlamydomonas _euryale	4	28	0.18	24	70	244	0.44
Chlamydomonas _euryale	4	28	0.18	24	500	238	0.42
Chlamydomonas _euryale	8	16	0.08	24	30	254	0.47
Chlamydomonas _euryale	8	16	0.09	24	70	238	0.48

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _euryale	8	16	0.09	24	500	233	0.46
Chlamydomonas _euryale	8	20	0.08	24	70	250	0.46
Chlamydomonas _euryale	8	20	0.09	24	30	247	0.45
Chlamydomonas _euryale	8	20	0.09	24	500	244	0.45
Chlamydomonas _euryale	8	24	0.09	24	70	242	0.64
Chlamydomonas _euryale	8	24	0.09	24	500	225	0.55
Chlamydomonas _euryale	8	24	0.09	24	30	230	0.43
Chlamydomonas _euryale	8	28	0.08	24	30	255	0.46
Chlamydomonas _euryale	8	28	0.09	24	70	237	0.46
Chlamydomonas _euryale	8	28	0.09	24	500	240	0.44
Chlamydomonas _euryale	16	16	0.04	24	30	239	0.47
Chlamydomonas _euryale	16	16	0.04	24	70	237	0.48
Chlamydomonas _euryale	16	16	0.05	24	500	236	0.46
Chlamydomonas _euryale	16	20	0.04	24	70	245	0.47
Chlamydomonas _euryale	16	20	0.04	24	500	243	0.46
Chlamydomonas _euryale	16	20	0.04	24	30	242	0.47
Chlamydomonas _euryale	16	24	0.05	24	70	240	0.65
Chlamydomonas _euryale	16	24	0.05	24	500	237	0.57

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _euryale	16	24	0.05	24	30	226	0.44
Chlamydomonas _euryale	16	28	0.04	24	30	248	0.47
Chlamydomonas _euryale	16	28	0.05	24	500	232	0.46
Chlamydomonas _euryale	16	28	0.05	24	70	233	0.48
Chlamydomonas _ICEMDV	1	4	0.74	4	70	227	0.65
Chlamydomonas _ICEMDV	1	4	0.74	4	70	224	0.61
Chlamydomonas _ICEMDV	1	8	0.72	4	70	227	0.61
Chlamydomonas _ICEMDV	1	8	0.72	4	70	239	0.60
Chlamydomonas _ICEMDV	1	12	0.68	4	70	234	0.63
Chlamydomonas _ICEMDV	1	12	0.68	4	70	258	0.63
Chlamydomonas _ICEMDV	2	4	0.36	4	70	229	0.66
Chlamydomonas _ICEMDV	2	4	0.36	4	70	229	0.61
Chlamydomonas _ICEMDV	2	8	0.35	4	70	228	0.62
Chlamydomonas _ICEMDV	2	8	0.35	4	70	241	0.60
Chlamydomonas _ICEMDV	2	12	0.34	4	70	236	0.64
Chlamydomonas _ICEMDV	2	12	0.34	4	70	261	0.64
Chlamydomonas _ICEMDV	4	4	0.18	4	70	231	0.66
Chlamydomonas _ICEMDV	4	4	0.18	4	70	232	0.61

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _ICEMDV	4	8	0.18	4	70	229	0.62
Chlamydomonas _ICEMDV	4	8	0.18	4	70	244	0.60
Chlamydomonas _ICEMDV	4	12	0.17	4	70	240	0.64
Chlamydomonas _ICEMDV	4	12	0.17	4	70	264	0.64
Chlamydomonas _ICEMDV	8	4	0.09	4	70	235	0.66
Chlamydomonas _ICEMDV	8	4	0.09	4	70	233	0.61
Chlamydomonas _ICEMDV	8	8	0.09	4	70	229	0.62
Chlamydomonas _ICEMDV	8	8	0.09	4	70	248	0.60
Chlamydomonas _ICEMDV	8	12	0.08	4	70	241	0.64
Chlamydomonas _ICEMDV	8	12	0.08	4	70	264	0.64
Chlamydomonas _ICEMDV	16	4	0.04	4	70	229	0.66
Chlamydomonas _ICEMDV	16	4	0.04	4	70	242	0.61
Chlamydomonas _ICEMDV	16	8	0.04	4	70	229	0.62
Chlamydomonas _ICEMDV	16	8	0.04	4	70	250	0.61
Chlamydomonas _ICEMDV	16	12	0.04	4	70	238	0.64
Chlamydomonas _ICEMDV	16	12	0.04	4	70	261	0.65
Chlamydomonas _klinobasis	1	2	0.75	4	10	225	0.73
Chlamydomonas _klinobasis	1	2	0.75	4	0	224	0.73

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _klinobasis	1	2	0.80	4	30	208	0.70
Chlamydomonas _klinobasis	1	4	0.74	4	0	226	0.78
Chlamydomonas _klinobasis	1	4	0.75	4	10	222	0.74
Chlamydomonas _klinobasis	1	4	0.80	4	30	208	0.71
Chlamydomonas _klinobasis	1	8	0.72	4	0	236	0.70
Chlamydomonas _klinobasis	1	8	0.73	4	10	228	0.72
Chlamydomonas _klinobasis	1	8	0.77	4	30	218	0.69
Chlamydomonas _klinobasis	1	12	0.70	4	0	241	0.70
Chlamydomonas _klinobasis	1	12	0.70	4	10	239	0.69
Chlamydomonas _klinobasis	1	12	0.74	4	30	224	0.67
Chlamydomonas _klinobasis	1	16	0.79	4	30	211	0.53
Chlamydomonas _klinobasis	2	2	0.38	4	10	220	0.73
Chlamydomonas _klinobasis	2	2	0.38	4	0	220	0.73
Chlamydomonas _klinobasis	2	2	0.41	4	30	204	0.71
Chlamydomonas _klinobasis	2	4	0.38	4	0	221	0.78
Chlamydomonas _klinobasis	2	4	0.38	4	10	220	0.74
Chlamydomonas _klinobasis	2	4	0.40	4	30	206	0.71
Chlamydomonas _klinobasis	2	8	0.36	4	0	233	0.70

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _klinobasis	2	8	0.37	4	10	228	0.71
Chlamydomonas _klinobasis	2	8	0.39	4	30	215	0.69
Chlamydomonas _klinobasis	4	2	0.19	4	0	220	0.73
Chlamydomonas _klinobasis	4	2	0.19	4	10	220	0.73
Chlamydomonas _klinobasis	4	2	0.20	4	30	207	0.71
Chlamydomonas _klinobasis	4	4	0.19	4	0	222	0.77
Chlamydomonas _klinobasis	4	4	0.19	4	10	222	0.74
Chlamydomonas _klinobasis	4	4	0.20	4	30	208	0.71
Chlamydomonas _klinobasis	4	8	0.18	4	0	231	0.70
Chlamydomonas _klinobasis	4	8	0.18	4	10	229	0.71
Chlamydomonas _klinobasis	4	8	0.19	4	30	218	0.69
Chlamydomonas _klinobasis	4	12	0.18	4	10	238	0.69
Chlamydomonas _klinobasis	4	12	0.18	4	0	238	0.70
Chlamydomonas _klinobasis	4	12	0.18	4	30	228	0.67
Chlamydomonas _klinobasis	4	16	0.19	4	30	219	0.53
Chlamydomonas _klinobasis	8	2	0.09	4	10	220	0.73
Chlamydomonas _klinobasis	8	2	0.10	4	0	220	0.72
Chlamydomonas _klinobasis	8	2	0.10	4	30	209	0.71

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _klinobasis	8	4	0.09	4	10	226	0.74
Chlamydomonas _klinobasis	8	4	0.09	4	0	224	0.77
Chlamydomonas _klinobasis	8	4	0.10	4	30	214	0.70
Chlamydomonas _klinobasis	8	8	0.09	4	0	234	0.69
Chlamydomonas _klinobasis	8	8	0.09	4	10	234	0.70
Chlamydomonas _klinobasis	8	8	0.09	4	30	226	0.69
Chlamydomonas _klinobasis	8	12	0.09	4	10	244	0.67
Chlamydomonas _klinobasis	8	12	0.09	4	0	236	0.69
Chlamydomonas _klinobasis	8	12	0.09	4	30	234	0.67
Chlamydomonas _klinobasis	8	16	0.09	4	30	221	0.53
Chlamydomonas _klinobasis	16	2	0.05	4	10	225	0.72
Chlamydomonas _klinobasis	16	2	0.05	4	0	222	0.72
Chlamydomonas _klinobasis	16	2	0.05	4	30	213	0.70
Chlamydomonas _klinobasis	16	4	0.05	4	10	227	0.73
Chlamydomonas _klinobasis	16	4	0.05	4	0	225	0.76
Chlamydomonas _klinobasis	16	4	0.05	4	30	218	0.70
Chlamydomonas _klinobasis	16	8	0.04	4	10	237	0.69
Chlamydomonas _klinobasis	16	8	0.04	4	0	233	0.69

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _klinobasis	16	8	0.05	4	30	227	0.69
Chlamydomonas _klinobasis	16	12	0.04	4	10	240	0.68
Chlamydomonas _klinobasis	16	12	0.04	4	30	236	0.68
Chlamydomonas _klinobasis	16	12	0.04	4	0	235	0.69
Chlamydomonas _klinobasis	16	16	0.05	4	30	221	0.53
Chlamydomonas _malina	1	2	0.64	4	500	258	0.57
Chlamydomonas _malina	1	2	0.74	4	70	222	0.63
Chlamydomonas _malina	1	2	0.78	4	30	211	0.69
Chlamydomonas _malina	1	4	0.65	4	70	253	0.66
Chlamydomonas _malina	1	4	0.65	4	70	259	0.62
Chlamydomonas _malina	1	4	0.66	4	500	254	0.56
Chlamydomonas _malina	1	4	0.72	4	30	229	0.70
Chlamydomonas _malina	1	4	0.74	4	70	225	0.72
Chlamydomonas _malina	1	8	0.59	4	500	282	0.55
Chlamydomonas _malina	1	8	0.66	4	70	244	0.62
Chlamydomonas _malina	1	8	0.66	4	70	263	0.60
Chlamydomonas _malina	1	8	0.70	4	30	242	0.65
Chlamydomonas _malina	1	8	0.70	4	70	239	0.68

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _malina	1	12	0.57	4	500	295	0.53
Chlamydomonas _malina	1	12	0.60	4	70	259	0.65
Chlamydomonas _malina	1	12	0.60	4	70	296	0.64
Chlamydomonas _malina	1	12	0.69	4	30	242	0.68
Chlamydomonas _malina	1	12	0.70	4	70	239	0.64
Chlamydomonas _malina	2	2	0.32	4	500	256	0.57
Chlamydomonas _malina	2	2	0.38	4	70	222	0.64
Chlamydomonas _malina	2	2	0.40	4	30	211	0.69
Chlamydomonas _malina	2	4	0.32	4	500	262	0.57
Chlamydomonas _malina	2	4	0.32	4	70	254	0.67
Chlamydomonas _malina	2	4	0.32	4	70	261	0.62
Chlamydomonas _malina	2	4	0.37	4	30	225	0.70
Chlamydomonas _malina	2	4	0.37	4	70	222	0.72
Chlamydomonas _malina	2	8	0.30	4	500	278	0.55
Chlamydomonas _malina	2	8	0.32	4	70	251	0.63
Chlamydomonas _malina	2	8	0.32	4	70	265	0.61
Chlamydomonas _malina	2	8	0.35	4	30	237	0.65
Chlamydomonas _malina	2	8	0.36	4	70	237	0.69

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _malina	2	12	0.30	4	70	263	0.65
Chlamydomonas _malina	2	12	0.30	4	70	298	0.65
Chlamydomonas _malina	4	2	0.16	4	500	255	0.57
Chlamydomonas _malina	4	2	0.19	4	70	213	0.64
Chlamydomonas _malina	4	2	0.20	4	30	210	0.69
Chlamydomonas _malina	4	4	0.16	4	70	254	0.66
Chlamydomonas _malina	4	4	0.16	4	70	262	0.62
Chlamydomonas _malina	4	4	0.16	4	500	254	0.57
Chlamydomonas _malina	4	4	0.19	4	30	221	0.70
Chlamydomonas _malina	4	4	0.19	4	70	224	0.71
Chlamydomonas _malina	4	8	0.15	4	500	278	0.55
Chlamydomonas _malina	4	8	0.16	4	70	255	0.64
Chlamydomonas _malina	4	8	0.16	4	70	265	0.61
Chlamydomonas _malina	4	8	0.18	4	70	236	0.70
Chlamydomonas _malina	4	8	0.18	4	30	235	0.65
Chlamydomonas _malina	4	12	0.15	4	500	287	0.53
Chlamydomonas _malina	4	12	0.15	4	70	265	0.65
Chlamydomonas _malina	4	12	0.15	4	70	300	0.65

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _malina	4	12	0.17	4	70	246	0.64
Chlamydomonas _malina	4	12	0.17	4	30	238	0.67
Chlamydomonas _malina	8	2	0.08	4	500	252	0.56
Chlamydomonas _malina	8	2	0.10	4	30	208	0.69
Chlamydomonas _malina	8	2	0.10	4	70	204	0.63
Chlamydomonas _malina	8	4	0.08	4	70	256	0.67
Chlamydomonas _malina	8	4	0.08	4	70	263	0.63
Chlamydomonas _malina	8	4	0.08	4	500	254	0.57
Chlamydomonas _malina	8	4	0.09	4	30	228	0.70
Chlamydomonas _malina	8	4	0.09	4	70	225	0.71
Chlamydomonas _malina	8	8	0.07	4	500	286	0.54
Chlamydomonas _malina	8	8	0.08	4	70	263	0.61
Chlamydomonas _malina	8	8	0.09	4	30	239	0.65
Chlamydomonas _malina	8	8	0.09	4	70	239	0.69
Chlamydomonas _malina	8	12	0.07	4	500	288	0.52
Chlamydomonas _malina	8	12	0.07	4	70	266	0.65
Chlamydomonas _malina	8	12	0.07	4	70	301	0.65
Chlamydomonas _malina	8	12	0.08	4	30	253	0.67

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _malina	8	12	0.09	4	70	243	0.65
Chlamydomonas _malina	16	2	0.05	4	30	210	0.68
Chlamydomonas _malina	16	2	0.05	4	70	210	0.63
Chlamydomonas _malina	16	4	0.04	4	70	255	0.67
Chlamydomonas _malina	16	4	0.04	4	70	266	0.65
Chlamydomonas _malina	16	4	0.04	4	500	267	0.56
Chlamydomonas _malina	16	4	0.04	4	30	232	0.69
Chlamydomonas _malina	16	4	0.05	4	70	231	0.71
Chlamydomonas _malina	16	8	0.04	4	500	290	0.53
Chlamydomonas _malina	16	8	0.04	4	70	246	0.65
Chlamydomonas _malina	16	8	0.04	4	70	259	0.63
Chlamydomonas _malina	16	8	0.04	4	30	246	0.65
Chlamydomonas _malina	16	8	0.04	4	70	245	0.69
Chlamydomonas _malina	16	12	0.03	4	500	305	0.52
Chlamydomonas _malina	16	12	0.04	4	70	267	0.65
Chlamydomonas _malina	16	12	0.04	4	70	300	0.64
Chlamydomonas _malina	16	12	0.04	4	30	260	0.67
Chlamydomonas _malina	16	12	0.04	4	70	248	0.65

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _priscui	1	2	0.68	4	500	245	0.59
Chlamydomonas _priscui	1	2	0.79	4	70	213	0.65
Chlamydomonas _priscui	1	2	0.82	4	30	203	0.69
Chlamydomonas _priscui	1	4	0.71	4	500	238	0.59
Chlamydomonas _priscui	1	4	0.77	4	70	209	0.64
Chlamydomonas _priscui	1	4	0.77	4	70	213	0.62
Chlamydomonas _priscui	1	4	0.77	4	700	224	0.45
Chlamydomonas _priscui	1	4	0.78	4	70	212	0.64
Chlamydomonas _priscui	1	4	0.81	4	30	204	0.70
Chlamydomonas _priscui	1	8	0.65	4	500	259	0.58
Chlamydomonas _priscui	1	8	0.70	4	70	240	0.65
Chlamydomonas _priscui	1	8	0.70	4	70	229	0.61
Chlamydomonas _priscui	1	8	0.70	4	700	249	0.45
Chlamydomonas _priscui	1	8	0.76	4	70	219	0.65
Chlamydomonas _priscui	1	8	0.76	4	30	220	0.65
Chlamydomonas _priscui	1	12	0.65	4	70	234	0.63
Chlamydomonas _priscui	1	12	0.65	4	70	263	0.63
Chlamydomonas _priscui	1	12	0.65	4	700	269	0.43

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _priscui	1	12	0.67	4	500	251	0.48
Chlamydomonas _priscui	1	12	0.79	4	30	213	0.64
Chlamydomonas _priscui	1	12	0.80	4	70	209	0.71
Chlamydomonas _priscui	1	16	0.65	4	500	257	0.47
Chlamydomonas _priscui	1	16	0.67	4	70	249	0.45
Chlamydomonas _priscui	1	16	0.79	4	30	210	0.52
Chlamydomonas _priscui	2	2	0.35	4	500	238	0.60
Chlamydomonas _priscui	2	2	0.40	4	70	202	0.65
Chlamydomonas _priscui	2	2	0.42	4	30	200	0.69
Chlamydomonas _priscui	2	4	0.34	4	500	250	0.58
Chlamydomonas _priscui	2	4	0.38	4	70	208	0.64
Chlamydomonas _priscui	2	4	0.38	4	70	215	0.63
Chlamydomonas _priscui	2	4	0.38	4	700	225	0.46
Chlamydomonas _priscui	2	4	0.40	4	70	212	0.64
Chlamydomonas _priscui	2	4	0.41	4	30	206	0.70
Chlamydomonas _priscui	2	8	0.33	4	500	252	0.58
Chlamydomonas _priscui	2	8	0.35	4	70	231	0.64
Chlamydomonas _priscui	2	8	0.35	4	70	231	0.61

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _priscui	2	8	0.35	4	700	250	0.45
Chlamydomonas _priscui	2	8	0.38	4	30	221	0.65
Chlamydomonas _priscui	2	8	0.38	4	70	216	0.66
Chlamydomonas _priscui	2	12	0.32	4	70	259	0.55
Chlamydomonas _priscui	2	12	0.33	4	70	235	0.63
Chlamydomonas _priscui	2	12	0.33	4	70	263	0.64
Chlamydomonas _priscui	2	12	0.33	4	700	268	0.43
Chlamydomonas _priscui	2	12	0.35	4	30	239	0.62
Chlamydomonas _priscui	2	12	0.37	4	500	227	0.56
Chlamydomonas _priscui	2	16	0.31	4	500	275	0.53
Chlamydomonas _priscui	2	16	0.36	4	70	235	0.58
Chlamydomonas _priscui	2	16	0.39	4	70	215	0.53
Chlamydomonas _priscui	2	16	0.39	4	30	211	0.51
Chlamydomonas _priscui	2	16	0.51	4	500	162	0.54
Chlamydomonas _priscui	4	2	0.18	4	500	237	0.60
Chlamydomonas _priscui	4	2	0.20	4	70	207	0.65
Chlamydomonas _priscui	4	2	0.21	4	30	199	0.69
Chlamydomonas _priscui	4	4	0.18	4	500	229	0.60

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _priscui	4	4	0.19	4	70	206	0.64
Chlamydomonas _priscui	4	4	0.19	4	70	215	0.63
Chlamydomonas _priscui	4	4	0.19	4	700	227	0.46
Chlamydomonas _priscui	4	4	0.20	4	70	211	0.64
Chlamydomonas _priscui	4	4	0.20	4	30	206	0.70
Chlamydomonas _priscui	4	8	0.16	4	500	247	0.58
Chlamydomonas _priscui	4	8	0.18	4	70	231	0.64
Chlamydomonas _priscui	4	8	0.18	4	70	231	0.61
Chlamydomonas _priscui	4	8	0.18	4	700	251	0.45
Chlamydomonas _priscui	4	8	0.19	4	30	225	0.65
Chlamydomonas _priscui	4	8	0.19	4	70	217	0.66
Chlamydomonas _priscui	4	12	0.16	4	70	236	0.62
Chlamydomonas _priscui	4	12	0.16	4	70	263	0.64
Chlamydomonas _priscui	4	12	0.16	4	700	268	0.43
Chlamydomonas _priscui	4	12	0.17	4	500	248	0.47
Chlamydomonas _priscui	4	12	0.19	4	30	218	0.63
Chlamydomonas _priscui	4	12	0.20	4	70	214	0.70
Chlamydomonas _priscui	4	16	0.16	4	500	257	0.47

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _priscui	4	16	0.17	4	70	254	0.46
Chlamydomonas _priscui	4	16	0.19	4	30	216	0.53
Chlamydomonas _priscui	8	2	0.09	4	500	231	0.60
Chlamydomonas _priscui	8	2	0.10	4	70	205	0.65
Chlamydomonas _priscui	8	2	0.10	4	30	203	0.69
Chlamydomonas _priscui	8	4	0.09	4	500	232	0.59
Chlamydomonas _priscui	8	4	0.10	4	70	208	0.64
Chlamydomonas _priscui	8	4	0.10	4	70	215	0.63
Chlamydomonas _priscui	8	4	0.10	4	700	230	0.47
Chlamydomonas _priscui	8	4	0.10	4	30	209	0.70
Chlamydomonas _priscui	8	4	0.10	4	70	208	0.64
Chlamydomonas _priscui	8	8	0.08	4	500	254	0.59
Chlamydomonas _priscui	8	8	0.09	4	70	232	0.64
Chlamydomonas _priscui	8	8	0.09	4	70	232	0.61
Chlamydomonas _priscui	8	8	0.09	4	700	251	0.45
Chlamydomonas _priscui	8	8	0.09	4	30	229	0.64
Chlamydomonas _priscui	8	8	0.10	4	70	219	0.66
Chlamydomonas _priscui	8	12	0.08	4	70	234	0.62

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _priscui	8	12	0.08	4	70	263	0.63
Chlamydomonas _priscui	8	12	0.08	4	700	267	0.43
Chlamydomonas _priscui	8	12	0.08	4	500	252	0.47
Chlamydomonas _priscui	8	12	0.09	4	30	224	0.62
Chlamydomonas _priscui	8	12	0.10	4	70	225	0.69
Chlamydomonas _priscui	8	16	0.08	4	500	254	0.47
Chlamydomonas _priscui	8	16	0.08	4	70	252	0.46
Chlamydomonas _priscui	8	16	0.09	4	30	223	0.52
Chlamydomonas _priscui	16	2	0.05	4	500	223	0.61
Chlamydomonas _priscui	16	2	0.05	4	70	204	0.64
Chlamydomonas _priscui	16	2	0.05	4	30	206	0.69
Chlamydomonas _priscui	16	4	0.05	4	70	220	0.63
Chlamydomonas _priscui	16	4	0.05	4	700	232	0.47
Chlamydomonas _priscui	16	4	0.05	4	500	224	0.61
Chlamydomonas _priscui	16	4	0.05	4	30	214	0.69
Chlamydomonas _priscui	16	4	0.05	4	70	215	0.64
Chlamydomonas _priscui	16	8	0.04	4	500	251	0.58
Chlamydomonas _priscui	16	8	0.04	4	70	231	0.64

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _priscui	16	8	0.04	4	70	222	0.63
Chlamydomonas _priscui	16	8	0.04	4	700	253	0.46
Chlamydomonas _priscui	16	8	0.04	4	30	236	0.64
Chlamydomonas _priscui	16	8	0.04	4	70	232	0.65
Chlamydomonas _priscui	16	12	0.04	4	500	249	0.47
Chlamydomonas _priscui	16	12	0.04	4	70	226	0.61
Chlamydomonas _priscui	16	12	0.04	4	70	257	0.63
Chlamydomonas _priscui	16	12	0.04	4	700	261	0.44
Chlamydomonas _priscui	16	12	0.05	4	70	232	0.70
Chlamydomonas _priscui	16	12	0.05	4	30	228	0.64
Chlamydomonas _priscui	16	16	0.04	4	500	245	0.47
Chlamydomonas _priscui	16	16	0.04	4	70	245	0.46
Chlamydomonas _priscui	16	16	0.05	4	30	224	0.52
Chlamydomonas _priscui	NA	4	NA	4	70	205	0.64
Chlamydomonas _reinhardtii	1	12	0.83	24	0	199	0.62
Chlamydomonas _reinhardtii	1	12	0.83	24	0	203	0.62
Chlamydomonas _reinhardtii	1	12	1.00	24	10	164	0.83
Chlamydomonas _reinhardtii	1	12	1.01	24	30	164	0.81

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _reinhardtii	1	12	1.03	24	0	163	0.80
Chlamydomonas _reinhardtii	1	16	0.78	24	0	222	0.62
Chlamydomonas _reinhardtii	1	16	0.78	24	0	209	0.61
Chlamydomonas _reinhardtii	1	16	0.96	24	0	161	0.58
Chlamydomonas _reinhardtii	1	16	1.06	24	10	157	0.58
Chlamydomonas _reinhardtii	1	16	1.07	24	30	155	0.58
Chlamydomonas _reinhardtii	1	20	0.81	24	0	216	0.59
Chlamydomonas _reinhardtii	1	20	0.81	24	0	200	0.57
Chlamydomonas _reinhardtii	1	20	1.03	24	0	164	0.56
Chlamydomonas _reinhardtii	1	20	1.04	24	10	162	0.57
Chlamydomonas _reinhardtii	1	20	1.05	24	30	159	0.56
Chlamydomonas _reinhardtii	1	24	0.84	24	0	197	0.55
Chlamydomonas _reinhardtii	1	24	0.84	24	0	202	0.55
Chlamydomonas _reinhardtii	1	24	0.92	24	10	167	0.56
Chlamydomonas _reinhardtii	1	24	1.02	24	30	163	0.54
Chlamydomonas _reinhardtii	1	24	1.03	24	0	161	0.55
Chlamydomonas _reinhardtii	1	28	0.96	24	30	173	0.50
Chlamydomonas _reinhardtii	1	28	1.00	24	0	168	0.53

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _reinhardtii	1	28	1.02	24	10	162	0.61
Chlamydomonas _reinhardtii	1	28	1.02	24	30	162	0.62
Chlamydomonas _reinhardtii	2	12	0.41	24	0	203	0.62
Chlamydomonas _reinhardtii	2	12	0.41	24	0	204	0.63
Chlamydomonas _reinhardtii	2	16	0.38	24	0	226	0.62
Chlamydomonas _reinhardtii	2	16	0.38	24	0	212	0.61
Chlamydomonas _reinhardtii	2	16	0.45	24	30	172	0.56
Chlamydomonas _reinhardtii	2	16	0.46	24	10	169	0.56
Chlamydomonas _reinhardtii	2	16	0.51	24	0	162	0.55
Chlamydomonas _reinhardtii	2	20	0.40	24	0	220	0.60
Chlamydomonas _reinhardtii	2	20	0.40	24	0	200	0.58
Chlamydomonas _reinhardtii	2	20	0.49	24	30	172	0.53
Chlamydomonas _reinhardtii	2	20	0.49	24	10	169	0.54
Chlamydomonas _reinhardtii	2	20	0.51	24	0	161	0.53
Chlamydomonas _reinhardtii	2	24	0.41	24	0	202	0.56
Chlamydomonas _reinhardtii	2	24	0.41	24	0	206	0.56
Chlamydomonas _reinhardtii	2	24	0.49	24	10	171	0.53
Chlamydomonas _reinhardtii	2	24	0.50	24	30	170	0.52

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _reinhardtii	2	24	0.50	24	0	166	0.53
Chlamydomonas _reinhardtii	2	28	0.48	24	30	178	0.50
Chlamydomonas _reinhardtii	2	28	0.49	24	10	170	0.51
Chlamydomonas _reinhardtii	2	28	0.50	24	0	168	0.49
Chlamydomonas _reinhardtii	4	12	0.20	24	0	206	0.62
Chlamydomonas _reinhardtii	4	12	0.20	24	0	209	0.63
Chlamydomonas _reinhardtii	4	12	0.25	24	10	169	0.84
Chlamydomonas _reinhardtii	4	12	0.25	24	30	166	0.82
Chlamydomonas _reinhardtii	4	12	0.26	24	0	162	0.82
Chlamydomonas _reinhardtii	4	16	0.19	24	0	229	0.63
Chlamydomonas _reinhardtii	4	16	0.19	24	0	219	0.62
Chlamydomonas _reinhardtii	4	16	0.26	24	0	164	0.57
Chlamydomonas _reinhardtii	4	16	0.27	24	10	158	0.58
Chlamydomonas _reinhardtii	4	16	0.27	24	30	155	0.57
Chlamydomonas _reinhardtii	4	20	0.20	24	0	227	0.60
Chlamydomonas _reinhardtii	4	20	0.20	24	0	198	0.58
Chlamydomonas _reinhardtii	4	20	0.24	24	0	165	0.55
Chlamydomonas _reinhardtii	4	20	0.25	24	10	168	0.57

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _reinhardtii	4	20	0.26	24	30	159	0.56
Chlamydomonas _reinhardtii	4	24	0.20	24	0	202	0.56
Chlamydomonas _reinhardtii	4	24	0.20	24	0	211	0.57
Chlamydomonas _reinhardtii	4	24	0.23	24	0	170	0.55
Chlamydomonas _reinhardtii	4	24	0.24	24	10	176	0.57
Chlamydomonas _reinhardtii	4	24	0.25	24	30	166	0.54
Chlamydomonas _reinhardtii	4	28	0.25	24	10	171	0.53
Chlamydomonas _reinhardtii	4	28	0.25	24	30	167	0.64
Chlamydomonas _reinhardtii	4	28	0.25	24	0	166	0.53
Chlamydomonas _reinhardtii	8	12	0.10	24	0	210	0.63
Chlamydomonas _reinhardtii	8	12	0.10	24	0	213	0.63
Chlamydomonas _reinhardtii	8	12	0.12	24	10	169	0.87
Chlamydomonas _reinhardtii	8	12	0.12	24	30	167	0.83
Chlamydomonas _reinhardtii	8	12	0.13	24	0	167	0.86
Chlamydomonas _reinhardtii	8	16	0.09	24	0	232	0.63
Chlamydomonas _reinhardtii	8	16	0.09	24	0	222	0.62
Chlamydomonas _reinhardtii	8	16	0.12	24	0	164	0.57
Chlamydomonas _reinhardtii	8	16	0.13	24	10	164	0.58

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _reinhardtii	8	16	0.13	24	30	157	0.57
Chlamydomonas _reinhardtii	8	20	0.10	24	0	226	0.61
Chlamydomonas _reinhardtii	8	20	0.10	24	0	195	0.59
Chlamydomonas _reinhardtii	8	20	0.12	24	10	170	0.57
Chlamydomonas _reinhardtii	8	20	0.12	24	0	171	0.56
Chlamydomonas _reinhardtii	8	20	0.13	24	30	162	0.56
Chlamydomonas _reinhardtii	8	24	0.10	24	0	197	0.57
Chlamydomonas _reinhardtii	8	24	0.10	24	0	204	0.58
Chlamydomonas _reinhardtii	8	24	0.12	24	10	171	0.56
Chlamydomonas _reinhardtii	8	24	0.13	24	30	166	0.55
Chlamydomonas _reinhardtii	8	24	0.13	24	0	167	0.54
Chlamydomonas _reinhardtii	8	28	0.12	24	10	174	0.54
Chlamydomonas _reinhardtii	8	28	0.12	24	30	169	0.62
Chlamydomonas _reinhardtii	8	28	0.13	24	0	167	0.54
Chlamydomonas _reinhardtii	16	12	0.05	24	0	214	0.64
Chlamydomonas _reinhardtii	16	12	0.05	24	0	216	0.64
Chlamydomonas _reinhardtii	16	12	0.06	24	0	159	0.98
Chlamydomonas _reinhardtii	16	12	0.06	24	30	170	0.84

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlamydomonas _reinhardtii	16	12	0.06	24	10	155	1.00
Chlamydomonas _reinhardtii	16	16	0.05	24	0	233	0.64
Chlamydomonas _reinhardtii	16	16	0.05	24	0	224	0.63
Chlamydomonas _reinhardtii	16	16	0.06	24	10	173	0.58
Chlamydomonas _reinhardtii	16	16	0.06	24	0	169	0.56
Chlamydomonas _reinhardtii	16	16	0.07	24	30	158	0.57
Chlamydomonas _reinhardtii	16	20	0.05	24	0	215	0.62
Chlamydomonas _reinhardtii	16	20	0.05	24	0	192	0.61
Chlamydomonas _reinhardtii	16	20	0.06	24	0	171	0.56
Chlamydomonas _reinhardtii	16	20	0.06	24	10	167	0.58
Chlamydomonas _reinhardtii	16	20	0.06	24	30	162	0.57
Chlamydomonas _reinhardtii	16	24	0.05	24	0	184	0.58
Chlamydomonas _reinhardtii	16	24	0.05	24	0	202	0.61
Chlamydomonas _reinhardtii	16	24	0.06	24	0	172	0.54
Chlamydomonas _reinhardtii	16	24	0.06	24	10	168	0.57
Chlamydomonas _reinhardtii	16	24	0.06	24	30	165	0.57
Chlamydomonas _reinhardtii	16	28	0.06	24	10	171	0.55
Chlamydomonas _reinhardtii	16	28	0.06	24	0	174	0.54

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
<i>Chlamydomonas</i> <i>_reinhardtii</i>	16	28	0.06	24	30	169	0.64
<i>Fragilariopsis_cy</i> <i>lindrus</i>	1	0	0.60	0	500	327	0.60
<i>Fragilariopsis_cy</i> <i>lindrus</i>	1	0	0.60	0	500	210	0.55
<i>Fragilariopsis_cy</i> <i>lindrus</i>	1	0	0.66	6	500	308	0.62
<i>Fragilariopsis_cy</i> <i>lindrus</i>	1	0	0.66	6	500	199	0.57
<i>Fragilariopsis_cy</i> <i>lindrus</i>	1	10	0.49	6	500	342	0.61
<i>Fragilariopsis_cy</i> <i>lindrus</i>	1	10	0.49	6	500	338	0.62
<i>Fragilariopsis_cy</i> <i>lindrus</i>	1	10	0.55	0	500	375	0.58
<i>Fragilariopsis_cy</i> <i>lindrus</i>	1	10	0.55	0	500	236	0.53
<i>Fragilariopsis_cy</i> <i>lindrus</i>	1	2	0.61	0	500	342	0.55
<i>Fragilariopsis_cy</i> <i>lindrus</i>	1	2	0.61	0	500	203	0.54
<i>Fragilariopsis_cy</i> <i>lindrus</i>	1	2	0.65	6	500	307	0.63
<i>Fragilariopsis_cy</i> <i>lindrus</i>	1	2	0.65	6	500	202	0.57
<i>Fragilariopsis_cy</i> <i>lindrus</i>	1	6	0.60	6	500	348	0.58
<i>Fragilariopsis_cy</i> <i>lindrus</i>	1	6	0.60	6	500	215	0.57
<i>Fragilariopsis_cy</i> <i>lindrus</i>	1	6	0.77	0	500	214	0.54
<i>Fragilariopsis_cy</i> <i>lindrus</i>	16	0	0.04	0	500	341	0.62
<i>Fragilariopsis_cy</i> <i>lindrus</i>	16	0	0.04	0	500	217	0.58

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Fragilariopsis_cy lindrus	16	0	0.04	6	500	317	0.65
Fragilariopsis_cy lindrus	16	0	0.04	6	500	204	0.60
Fragilariopsis_cy lindrus	16	10	0.03	6	500	357	0.61
Fragilariopsis_cy lindrus	16	10	0.03	6	500	346	0.62
Fragilariopsis_cy lindrus	16	10	0.03	0	500	388	0.58
Fragilariopsis_cy lindrus	16	10	0.03	0	500	245	0.55
Fragilariopsis_cy lindrus	16	2	0.03	0	500	344	0.56
Fragilariopsis_cy lindrus	16	2	0.03	0	500	346	0.55
Fragilariopsis_cy lindrus	16	2	0.03	0	500	210	0.56
Fragilariopsis_cy lindrus	16	2	0.04	6	500	322	0.64
Fragilariopsis_cy lindrus	16	2	0.04	6	500	214	0.58
Fragilariopsis_cy lindrus	16	6	0.04	6	500	356	0.59
Fragilariopsis_cy lindrus	16	6	0.04	6	500	228	0.59
Fragilariopsis_cy lindrus	16	6	0.05	0	500	223	0.56
Fragilariopsis_cy lindrus	2	0	0.30	0	500	329	0.60
Fragilariopsis_cy lindrus	2	0	0.30	0	500	211	0.56
Fragilariopsis_cy lindrus	2	0	0.33	6	500	307	0.63
Fragilariopsis_cy lindrus	2	0	0.33	6	500	198	0.58

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Fragilariopsis_cy lindrus	2	10	0.24	6	500	344	0.62
Fragilariopsis_cy lindrus	2	10	0.24	6	500	339	0.62
Fragilariopsis_cy lindrus	2	10	0.27	0	500	376	0.58
Fragilariopsis_cy lindrus	2	10	0.27	0	500	238	0.54
Fragilariopsis_cy lindrus	2	2	0.31	0	500	338	0.55
Fragilariopsis_cy lindrus	2	2	0.31	0	500	203	0.55
Fragilariopsis_cy lindrus	2	2	0.32	6	500	311	0.64
Fragilariopsis_cy lindrus	2	2	0.32	6	500	202	0.57
Fragilariopsis_cy lindrus	2	6	0.30	6	500	347	0.58
Fragilariopsis_cy lindrus	2	6	0.30	6	500	215	0.57
Fragilariopsis_cy lindrus	2	6	0.38	0	500	215	0.54
Fragilariopsis_cy lindrus	4	0	0.15	0	500	331	0.61
Fragilariopsis_cy lindrus	4	0	0.15	0	500	213	0.56
Fragilariopsis_cy lindrus	4	0	0.16	6	500	308	0.64
Fragilariopsis_cy lindrus	4	0	0.16	6	500	199	0.58
Fragilariopsis_cy lindrus	4	10	0.12	6	500	347	0.62
Fragilariopsis_cy lindrus	4	10	0.12	6	500	341	0.63
Fragilariopsis_cy lindrus	4	10	0.13	0	500	379	0.59

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Fragilariopsis_cy lindrus	4	10	0.13	0	500	241	0.54
Fragilariopsis_cy lindrus	4	2	0.15	0	500	333	0.56
Fragilariopsis_cy lindrus	4	2	0.15	0	500	206	0.55
Fragilariopsis_cy lindrus	4	2	0.16	6	500	308	0.64
Fragilariopsis_cy lindrus	4	2	0.16	6	500	203	0.58
Fragilariopsis_cy lindrus	4	6	0.15	6	500	349	0.59
Fragilariopsis_cy lindrus	4	6	0.15	6	500	217	0.58
Fragilariopsis_cy lindrus	4	6	0.19	0	500	217	0.55
Fragilariopsis_cy lindrus	8	0	0.07	0	500	337	0.61
Fragilariopsis_cy lindrus	8	0	0.07	0	500	215	0.57
Fragilariopsis_cy lindrus	8	0	0.08	6	500	312	0.64
Fragilariopsis_cy lindrus	8	0	0.08	6	500	201	0.59
Fragilariopsis_cy lindrus	8	10	0.06	6	500	352	0.62
Fragilariopsis_cy lindrus	8	10	0.06	6	500	343	0.63
Fragilariopsis_cy lindrus	8	10	0.07	0	500	384	0.58
Fragilariopsis_cy lindrus	8	10	0.07	0	500	245	0.55
Fragilariopsis_cy lindrus	8	2	0.07	0	500	349	0.55
Fragilariopsis_cy lindrus	8	2	0.07	0	500	209	0.56

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Fragilariopsis_cy lindrus	8	2	0.08	6	500	310	0.64
Fragilariopsis_cy lindrus	8	2	0.08	6	500	207	0.59
Fragilariopsis_cy lindrus	8	6	0.07	6	500	352	0.59
Fragilariopsis_cy lindrus	8	6	0.07	6	500	221	0.58
Fragilariopsis_cy lindrus	8	6	0.09	0	500	222	0.55
Chlorella_vulgari s	1	10	0.90	22	0	203	0.67
Chlorella_vulgari s	1	10	0.90	22	0	178	0.71
Chlorella_vulgari s	1	10	0.90	22	0	181	0.64
Chlorella_vulgari s	1	14	0.85	22	0	211	0.66
Chlorella_vulgari s	1	14	0.85	22	0	184	0.71
Chlorella_vulgari s	1	14	0.85	22	0	199	0.67
Chlorella_vulgari s	1	18	0.90	22	0	214	0.65
Chlorella_vulgari s	1	18	0.90	22	0	185	0.70
Chlorella_vulgari s	1	18	0.90	22	0	165	0.65
Chlorella_vulgari s	1	22	0.85	22	0	222	0.65
Chlorella_vulgari s	1	22	0.85	22	0	197	0.71
Chlorella_vulgari s	1	22	0.85	22	0	171	0.64
Chlorella_vulgari s	1	26	0.81	22	0	194	0.62

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlorella_vulgaris	1	26	0.81	22	0	209	0.63
Chlorella_vulgaris	16	10	0.05	22	0	209	0.68
Chlorella_vulgaris	16	10	0.05	22	0	191	0.72
Chlorella_vulgaris	16	10	0.05	22	0	202	0.66
Chlorella_vulgaris	16	14	0.05	22	0	213	0.67
Chlorella_vulgaris	16	14	0.05	22	0	200	0.73
Chlorella_vulgaris	16	14	0.05	22	0	209	0.66
Chlorella_vulgaris	16	18	0.05	22	0	218	0.66
Chlorella_vulgaris	16	18	0.05	22	0	202	0.73
Chlorella_vulgaris	16	18	0.05	22	0	183	0.66
Chlorella_vulgaris	16	22	0.05	22	0	222	0.66
Chlorella_vulgaris	16	22	0.05	22	0	202	0.73
Chlorella_vulgaris	16	22	0.05	22	0	187	0.66
Chlorella_vulgaris	16	26	0.05	22	0	200	0.74
Chlorella_vulgaris	16	26	0.05	22	0	195	0.64
Chlorella_vulgaris	16	26	0.05	22	0	215	0.67
Chlorella_vulgaris	2	10	0.44	22	0	203	0.67
Chlorella_vulgaris	2	10	0.44	22	0	180	0.72

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlorella_vulgaris	2	10	0.44	22	0	184	0.65
Chlorella_vulgaris	2	14	0.42	22	0	211	0.67
Chlorella_vulgaris	2	14	0.42	22	0	186	0.71
Chlorella_vulgaris	2	14	0.42	22	0	200	0.67
Chlorella_vulgaris	2	18	0.44	22	0	217	0.66
Chlorella_vulgaris	2	18	0.44	22	0	190	0.70
Chlorella_vulgaris	2	18	0.44	22	0	165	0.65
Chlorella_vulgaris	2	22	0.42	22	0	224	0.65
Chlorella_vulgaris	2	22	0.42	22	0	201	0.71
Chlorella_vulgaris	2	22	0.42	22	0	178	0.65
Chlorella_vulgaris	2	26	0.41	22	0	204	0.69
Chlorella_vulgaris	2	26	0.41	22	0	196	0.62
Chlorella_vulgaris	2	26	0.41	22	0	210	0.63
Chlorella_vulgaris	4	10	0.22	22	0	204	0.67
Chlorella_vulgaris	4	10	0.22	22	0	183	0.72
Chlorella_vulgaris	4	10	0.22	22	0	187	0.66
Chlorella_vulgaris	4	14	0.21	22	0	212	0.66
Chlorella_vulgaris	4	14	0.21	22	0	190	0.72

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlorella_vulgaris	4	14	0.21	22	0	202	0.67
Chlorella_vulgaris	4	18	0.21	22	0	218	0.66
Chlorella_vulgaris	4	18	0.21	22	0	198	0.71
Chlorella_vulgaris	4	18	0.21	22	0	172	0.66
Chlorella_vulgaris	4	22	0.21	22	0	225	0.64
Chlorella_vulgaris	4	22	0.21	22	0	202	0.70
Chlorella_vulgaris	4	22	0.21	22	0	182	0.65
Chlorella_vulgaris	4	26	0.20	22	0	208	0.70
Chlorella_vulgaris	4	26	0.20	22	0	198	0.62
Chlorella_vulgaris	4	26	0.20	22	0	217	0.63
Chlorella_vulgaris	8	10	0.11	22	0	206	0.67
Chlorella_vulgaris	8	10	0.11	22	0	187	0.72
Chlorella_vulgaris	8	10	0.11	22	0	194	0.66
Chlorella_vulgaris	8	14	0.10	22	0	211	0.66
Chlorella_vulgaris	8	14	0.10	22	0	195	0.72
Chlorella_vulgaris	8	14	0.10	22	0	207	0.67
Chlorella_vulgaris	8	18	0.11	22	0	218	0.65
Chlorella_vulgaris	8	18	0.11	22	0	203	0.72

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Chlorella_vulgari s	8	18	0.11	22	0	179	0.66
Chlorella_vulgari s	8	22	0.10	22	0	227	0.64
Chlorella_vulgari s	8	22	0.10	22	0	205	0.71
Chlorella_vulgari s	8	22	0.10	22	0	186	0.65
Chlorella_vulgari s	8	26	0.10	22	0	206	0.71
Chlorella_vulgari s	8	26	0.10	22	0	197	0.62
Chlorella_vulgari s	8	26	0.10	22	0	219	0.65
Thalassiosira_ps eudonana	1	10	0.53	22	0	282	0.58
Thalassiosira_ps eudonana	1	10	0.53	22	0	305	0.71
Thalassiosira_ps eudonana	1	10	0.53	22	0	348	0.67
Thalassiosira_ps eudonana	1	14	0.44	22	0	228	0.61
Thalassiosira_ps eudonana	1	14	0.44	22	0	611	0.65
Thalassiosira_ps eudonana	1	14	0.44	22	0	354	0.65
Thalassiosira_ps eudonana	1	14	0.44	22	0	343	0.61
Thalassiosira_ps eudonana	1	18	0.42	22	0	306	0.56
Thalassiosira_ps eudonana	1	18	0.42	22	0	639	0.65
Thalassiosira_ps eudonana	1	18	0.42	22	0	330	0.60
Thalassiosira_ps eudonana	1	18	0.42	22	0	338	0.60

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Thalassiosira_ps eudonana	1	20	0.39	22	0	645	0.64
Thalassiosira_ps eudonana	1	20	0.39	22	0	384	0.67
Thalassiosira_ps eudonana	1	20	0.39	22	0	353	0.58
Thalassiosira_ps eudonana	1	20	0.39	22	0	336	0.59
Thalassiosira_ps eudonana	1	22	0.52	22	0	371	0.77
Thalassiosira_ps eudonana	1	22	0.52	22	0	247	0.59
Thalassiosira_ps eudonana	1	22	0.52	22	0	356	0.58
Thalassiosira_ps eudonana	1	22	0.52	22	0	315	0.58
Thalassiosira_ps eudonana	1	24	0.38	22	0	655	0.63
Thalassiosira_ps eudonana	1	24	0.38	22	0	387	0.59
Thalassiosira_ps eudonana	1	24	0.38	22	0	360	0.57
Thalassiosira_ps eudonana	1	24	0.38	22	0	342	0.57
Thalassiosira_ps eudonana	1	28	0.42	22	0	500	0.53
Thalassiosira_ps eudonana	1	28	0.42	22	0	391	0.57
Thalassiosira_ps eudonana	1	28	0.42	22	0	353	0.55
Thalassiosira_ps eudonana	1	28	0.42	22	0	346	0.55
Thalassiosira_ps eudonana	16	10	0.03	22	0	301	0.70
Thalassiosira_ps eudonana	16	10	0.03	22	0	353	0.66

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Thalassiosira_ps eudonana	16	10	0.03	22	0	336	0.61
Thalassiosira_ps eudonana	16	10	0.03	22	0	319	0.63
Thalassiosira_ps eudonana	16	14	0.03	22	0	621	0.65
Thalassiosira_ps eudonana	16	14	0.03	22	0	357	0.65
Thalassiosira_ps eudonana	16	14	0.03	22	0	346	0.61
Thalassiosira_ps eudonana	16	14	0.03	22	0	330	0.62
Thalassiosira_ps eudonana	16	18	0.03	22	0	624	0.65
Thalassiosira_ps eudonana	16	18	0.03	22	0	317	0.61
Thalassiosira_ps eudonana	16	18	0.03	22	0	352	0.60
Thalassiosira_ps eudonana	16	18	0.03	22	0	330	0.61
Thalassiosira_ps eudonana	16	20	0.02	22	0	638	0.64
Thalassiosira_ps eudonana	16	20	0.02	22	0	374	0.62
Thalassiosira_ps eudonana	16	20	0.02	22	0	353	0.59
Thalassiosira_ps eudonana	16	20	0.02	22	0	334	0.61
Thalassiosira_ps eudonana	16	22	0.03	22	0	356	0.59
Thalassiosira_ps eudonana	16	22	0.03	22	0	335	0.60
Thalassiosira_ps eudonana	16	24	0.03	22	0	543	0.62
Thalassiosira_ps eudonana	16	24	0.03	22	0	375	0.62

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Thalassiosira_ps eudonana	16	24	0.03	22	0	356	0.58
Thalassiosira_ps eudonana	16	24	0.03	22	0	332	0.59
Thalassiosira_ps eudonana	16	28	0.02	22	0	662	0.60
Thalassiosira_ps eudonana	16	28	0.02	22	0	370	0.60
Thalassiosira_ps eudonana	16	28	0.02	22	0	351	0.57
Thalassiosira_ps eudonana	16	28	0.02	22	0	328	0.58
Thalassiosira_ps eudonana	2	10	0.27	22	0	289	0.58
Thalassiosira_ps eudonana	2	10	0.27	22	0	308	0.71
Thalassiosira_ps eudonana	2	10	0.27	22	0	349	0.67
Thalassiosira_ps eudonana	2	14	0.21	22	0	229	0.61
Thalassiosira_ps eudonana	2	14	0.21	22	0	633	0.65
Thalassiosira_ps eudonana	2	14	0.21	22	0	356	0.65
Thalassiosira_ps eudonana	2	14	0.21	22	0	347	0.60
Thalassiosira_ps eudonana	2	18	0.21	22	0	305	0.56
Thalassiosira_ps eudonana	2	18	0.21	22	0	633	0.65
Thalassiosira_ps eudonana	2	18	0.21	22	0	329	0.60
Thalassiosira_ps eudonana	2	18	0.21	22	0	338	0.60
Thalassiosira_ps eudonana	2	20	0.19	22	0	643	0.64

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Thalassiosira_ps eudonana	2	20	0.19	22	0	386	0.67
Thalassiosira_ps eudonana	2	20	0.19	22	0	356	0.58
Thalassiosira_ps eudonana	2	20	0.19	22	0	340	0.59
Thalassiosira_ps eudonana	2	22	0.26	22	0	246	0.59
Thalassiosira_ps eudonana	2	22	0.26	22	0	361	0.57
Thalassiosira_ps eudonana	2	22	0.26	22	0	343	0.58
Thalassiosira_ps eudonana	2	24	0.19	22	0	654	0.63
Thalassiosira_ps eudonana	2	24	0.19	22	0	389	0.59
Thalassiosira_ps eudonana	2	24	0.19	22	0	364	0.57
Thalassiosira_ps eudonana	2	24	0.19	22	0	344	0.57
Thalassiosira_ps eudonana	2	28	0.19	22	0	669	0.60
Thalassiosira_ps eudonana	2	28	0.19	22	0	394	0.57
Thalassiosira_ps eudonana	2	28	0.19	22	0	365	0.54
Thalassiosira_ps eudonana	2	28	0.19	22	0	346	0.55
Thalassiosira_ps eudonana	4	10	0.13	22	0	279	0.58
Thalassiosira_ps eudonana	4	10	0.13	22	0	300	0.70
Thalassiosira_ps eudonana	4	10	0.13	22	0	349	0.67
Thalassiosira_ps eudonana	4	14	0.11	22	0	229	0.61

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Thalassiosira_ps eudonana	4	14	0.11	22	0	625	0.65
Thalassiosira_ps eudonana	4	14	0.11	22	0	357	0.65
Thalassiosira_ps eudonana	4	14	0.11	22	0	347	0.61
Thalassiosira_ps eudonana	4	18	0.10	22	0	304	0.57
Thalassiosira_ps eudonana	4	18	0.10	22	0	630	0.65
Thalassiosira_ps eudonana	4	18	0.10	22	0	328	0.60
Thalassiosira_ps eudonana	4	18	0.10	22	0	339	0.60
Thalassiosira_ps eudonana	4	20	0.10	22	0	641	0.64
Thalassiosira_ps eudonana	4	20	0.10	22	0	386	0.67
Thalassiosira_ps eudonana	4	20	0.10	22	0	358	0.58
Thalassiosira_ps eudonana	4	20	0.10	22	0	339	0.59
Thalassiosira_ps eudonana	4	22	0.13	22	0	368	0.76
Thalassiosira_ps eudonana	4	22	0.13	22	0	248	0.59
Thalassiosira_ps eudonana	4	22	0.13	22	0	362	0.57
Thalassiosira_ps eudonana	4	22	0.13	22	0	343	0.58
Thalassiosira_ps eudonana	4	24	0.10	22	0	624	0.63
Thalassiosira_ps eudonana	4	24	0.10	22	0	387	0.59
Thalassiosira_ps eudonana	4	24	0.10	22	0	365	0.57

Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Thalassiosira_ps eudonana	4	24	0.10	22	0	343	0.57
Thalassiosira_ps eudonana	4	28	0.09	22	0	662	0.60
Thalassiosira_ps eudonana	4	28	0.09	22	0	389	0.57
Thalassiosira_ps eudonana	4	28	0.09	22	0	364	0.54
Thalassiosira_ps eudonana	4	28	0.09	22	0	349	0.55
Thalassiosira_ps eudonana	8	10	0.07	22	0	292	0.58
Thalassiosira_ps eudonana	8	10	0.07	22	0	302	0.70
Thalassiosira_ps eudonana	8	10	0.07	22	0	350	0.66
Thalassiosira_ps eudonana	8	14	0.05	22	0	229	0.61
Thalassiosira_ps eudonana	8	14	0.05	22	0	619	0.65
Thalassiosira_ps eudonana	8	14	0.05	22	0	356	0.65
Thalassiosira_ps eudonana	8	14	0.05	22	0	348	0.60
Thalassiosira_ps eudonana	8	18	0.05	22	0	302	0.57
Thalassiosira_ps eudonana	8	18	0.05	22	0	624	0.65
Thalassiosira_ps eudonana	8	18	0.05	22	0	327	0.60
Thalassiosira_ps eudonana	8	18	0.05	22	0	339	0.61
Thalassiosira_ps eudonana	8	20	0.05	22	0	636	0.64
Thalassiosira_ps eudonana	8	20	0.05	22	0	383	0.61

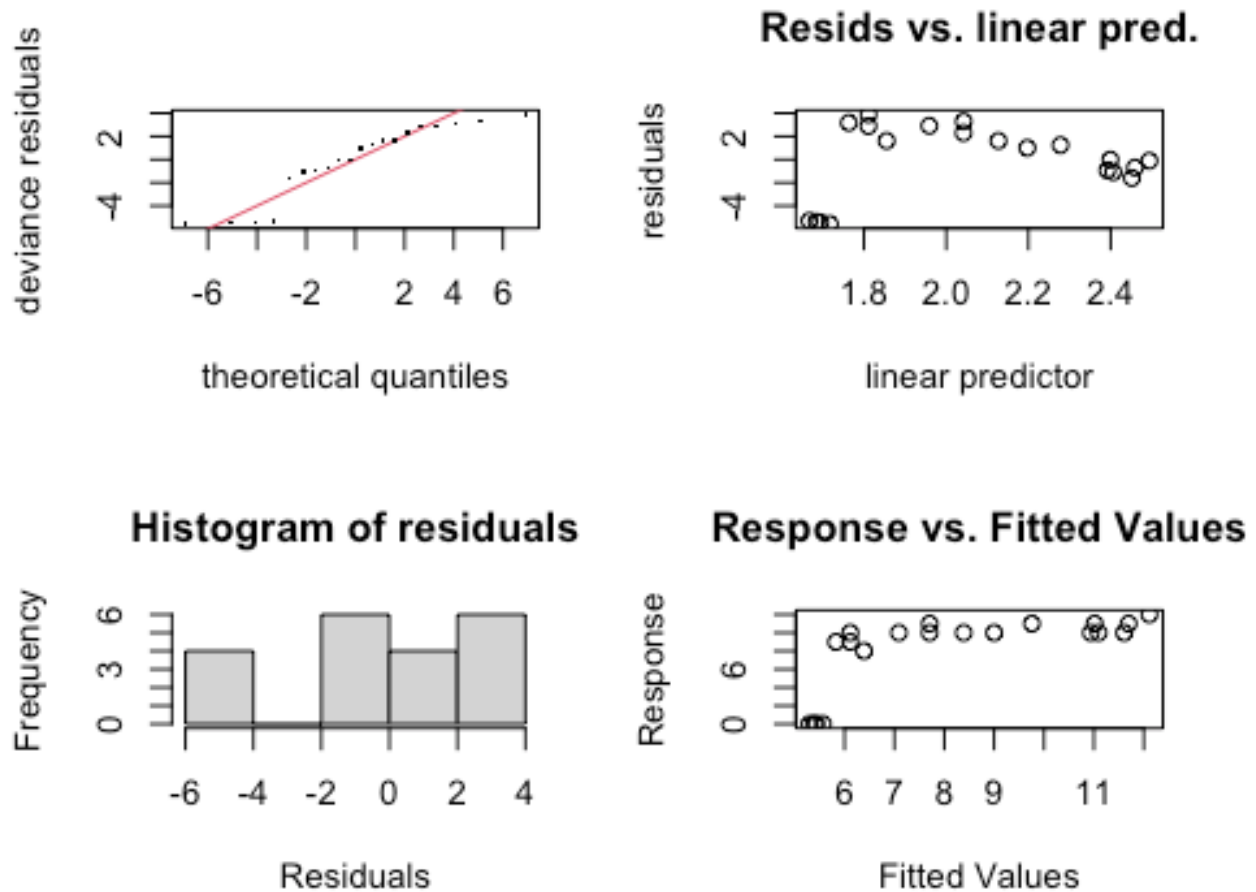
Taxa	Flash Spacing (s)	Measurement Temp (°C)	Equivalen t Light	Growth Temp (°C)	NaCl (mM)	σ_{PSI} I	FvF M
Thalassiosira_ps eudonana	8	20	0.05	22	0	357	0.58
Thalassiosira_ps eudonana	8	20	0.05	22	0	339	0.60
Thalassiosira_ps eudonana	8	22	0.06	22	0	365	0.76
Thalassiosira_ps eudonana	8	22	0.06	22	0	245	0.60
Thalassiosira_ps eudonana	8	22	0.06	22	0	361	0.58
Thalassiosira_ps eudonana	8	22	0.06	22	0	340	0.59
Thalassiosira_ps eudonana	8	24	0.05	22	0	603	0.63
Thalassiosira_ps eudonana	8	24	0.05	22	0	385	0.60
Thalassiosira_ps eudonana	8	24	0.05	22	0	363	0.57
Thalassiosira_ps eudonana	8	24	0.05	22	0	340	0.58
Thalassiosira_ps eudonana	8	28	0.05	22	0	662	0.60
Thalassiosira_ps eudonana	8	28	0.05	22	0	385	0.58
Thalassiosira_ps eudonana	8	28	0.05	22	0	358	0.55
Thalassiosira_ps eudonana	8	28	0.05	22	0	342	0.56

Table 5: Summary statistics by phytoplankton strain of GAM models using the restricted maximum likelihood method to model the response of the damping of SState induced chlorophyll fluorescence oscillations to the predictors of measurement temperature (°C) and the effective light level ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$).

Strain	term	edf	ref.df	statistic	p.value
Fragilariopsis_cylindrus	s(LightLevel_ue)	1.8 3	1.97	4.71	0.03
Fragilariopsis_cylindrus	s(Temp_C)	1.0 0	1.00	0.06	0.82
Fragilariopsis_cylindrus	ti(Temp_C,LightLevel_ue)	1.0 0	1.00	0.03	0.86
Fragilariopsis_cylindrus	s(LightLevel_ue)	1.1 0	1.20	3.27	0.07
Fragilariopsis_cylindrus	s(Temp_C)	1.0 0	1.00	2.36	0.15
Fragilariopsis_cylindrus	ti(Temp_C,LightLevel_ue)	1.4 6	1.70	2.82	0.18
Chlorella_vulgaris	s(LightLevel_ue)	1.9 4	2.00	8.41	0.00
Chlorella_vulgaris	s(Temp_C)	1.8 9	1.99	4.71	0.02
Chlorella_vulgaris	ti(Temp_C,LightLevel_ue)	1.0 0	1.00	0.71	0.41
Thalassiosira_pseudonana	s(LightLevel_ue)	1.9 6	1.99	10.25	0.00
Thalassiosira_pseudonana	s(Temp_C)	1.0 0	1.00	2.83	0.10
Thalassiosira_pseudonana	ti(Temp_C,LightLevel_ue)	2.1 7	2.63	3.96	0.04
Chlamydomonas_euryale	s(LightLevel_ue)	1.0 0	1.00	188.81	0.00
Chlamydomonas_euryale	s(Temp_C)	1.9 6	1.00	537.24	0.00
Chlamydomonas_euryale	ti(Temp_C,LightLevel_ue)	1.0 0	1.00	42.56	0.00
Chlamydomonas_ICEMDV	s(LightLevel_ue)	1.7 4	1.93	3.33	0.07
Chlamydomonas_ICEMDV	s(Temp_C)	1.0 0	1.00	0.30	0.59
Chlamydomonas_ICEMDV	ti(Temp_C,LightLevel_ue)	1.0 0	1.00	0.03	0.87
Chlamydomonas_klinobasis	s(LightLevel_ue)	1.8 5	1.98	5.81	0.01

Strain	term	edf	ref.df	statistic	p.value
Chlamydomonas_klinobasis	s(Temp_C)	1.2 5	1.44	2.07	0.15
Chlamydomonas_klinobasis	ti(Temp_C,LightLevel_ue)	1.0 0	1.00	2.76	0.12
Chlamydomonas_malina	s(LightLevel_ue)	1.8 4	1.97	3.74	0.06
Chlamydomonas_malina	s(Temp_C)	1.0 0	1.00	0.54	0.48
Chlamydomonas_malina	ti(Temp_C,LightLevel_ue)	1.0 0	1.00	0.99	0.34
Chlamydomonas_priscui	s(LightLevel_ue)	1.9 0	1.99	11.07	0.00
Chlamydomonas_priscui	s(Temp_C)	1.7 7	1.95	7.21	0.00
Chlamydomonas_priscui	ti(Temp_C,LightLevel_ue)	1.0 0	1.00	7.78	0.01
Chlamydomonas_reinhardt i	s(LightLevel_ue)	1.8 7	1.93	2.55	0.11
Chlamydomonas_reinhardt i	s(Temp_C)	1.0 0	1.00	6.54	0.01
Chlamydomonas_reinhardt i	ti(Temp_C,LightLevel_ue)	2.1 9	2.65	1.08	0.48

Strain: *Fragilariopsis_cylindrus*, GrowthTemp (°C): 0 °C



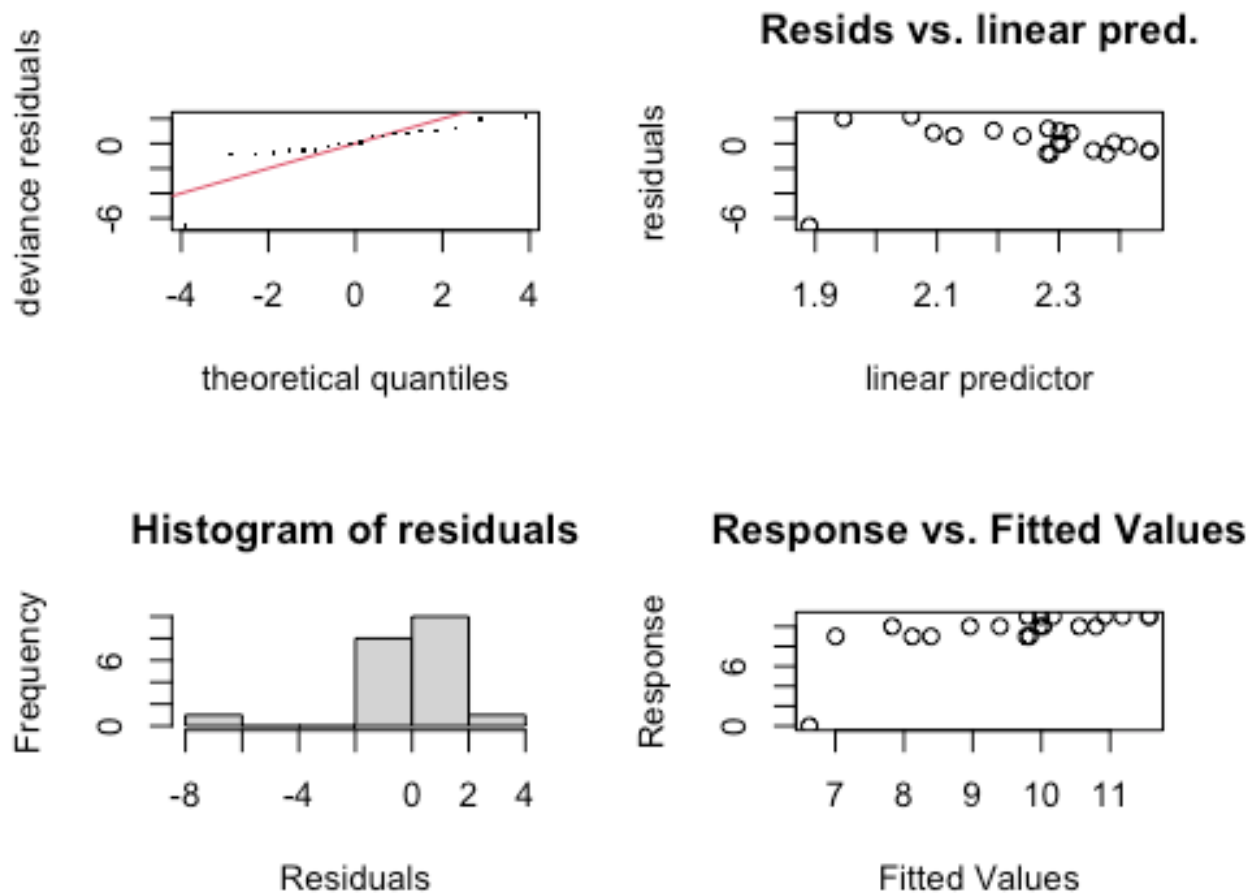
Method: REML Optimizer: outer newton full convergence after 12 iterations. Gradient range [-1.148855e-05,2.458705e-06] (score 56.65425 & scale 12.43105). Hessian positive definite, eigenvalue range [4.616448e-06,8.015894]. Model rank = 9 / 9

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

k'	edf	k-index	p-value
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$s(\text{LightLevel_ue})$ 2.00 1.83 0.26 $<2\text{e-}16$ * $s(\text{Temp_C})$ 2.00 1.00 0.81 0.145
 $ti(\text{Temp_C}, \text{LightLevel_ue})$ 4.00 1.00 0.55 0.005 — Signif. codes: 0 “0.001” 0.01 “0.05”
 0.1 “1”

Strain: *Fragilariopsis_cylindrus*, GrowthTemp (°C): 6°C



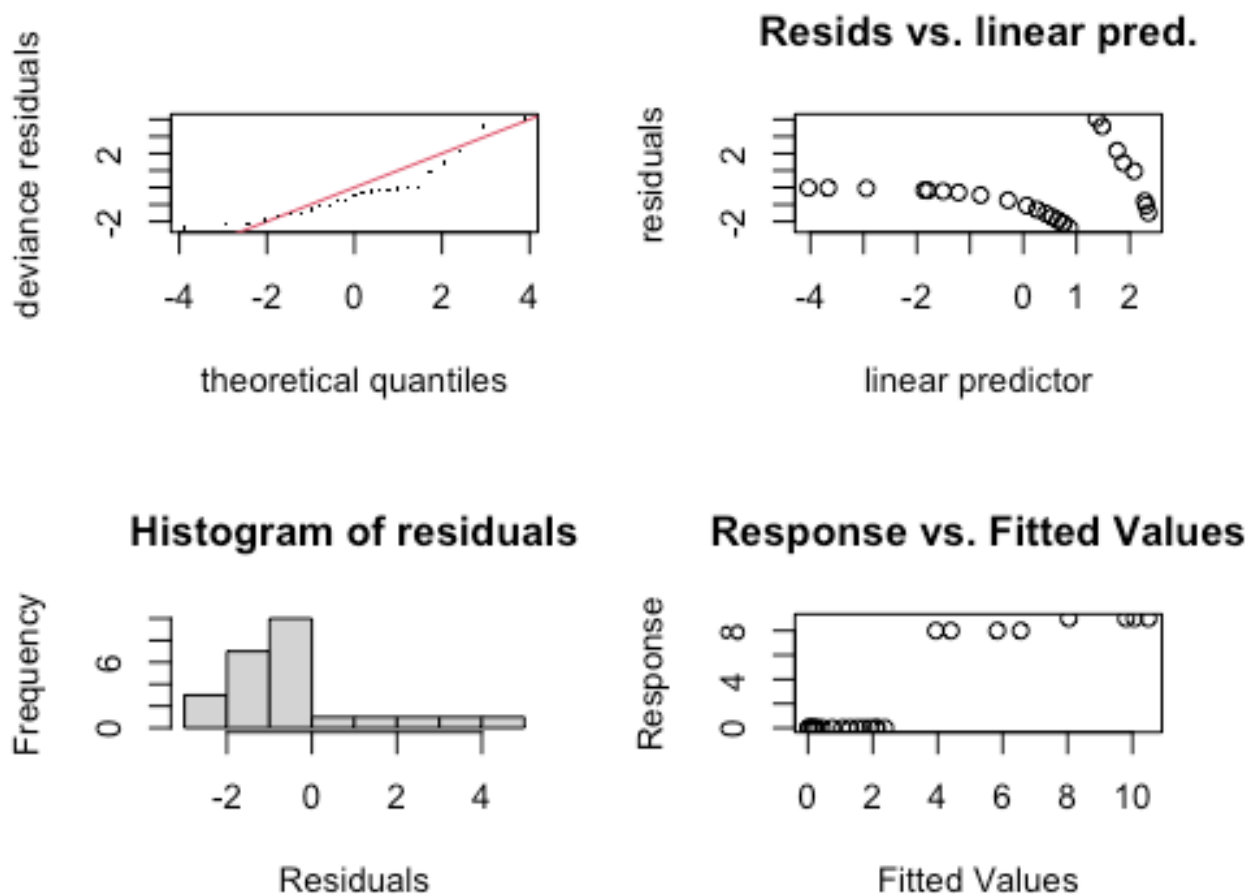
Method: REML Optimizer: outer newton full convergence after 12 iterations. Gradient range
 [-1.045605e-05, 1.128619e-05] (score 47.79832 & scale 3.965611). Hessian positive
 definite, eigenvalue range [6.856764e-06, 8.007953]. Model rank = 9 / 9

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value
s(LightLevel_ue)	2.00	1.10	0.76	0.055
s(Temp_C)	2.00	1.00	0.93	0.190
ti(Temp_C,LightLevel_ue)	4.00	1.46	0.89	0.070

— Signif. codes: 0 “**0.001**” 0.01 “0.05” 0.1 “1”

Strain: *Chlorella_vulgaris*, GrowthTemp (°C): 22°C



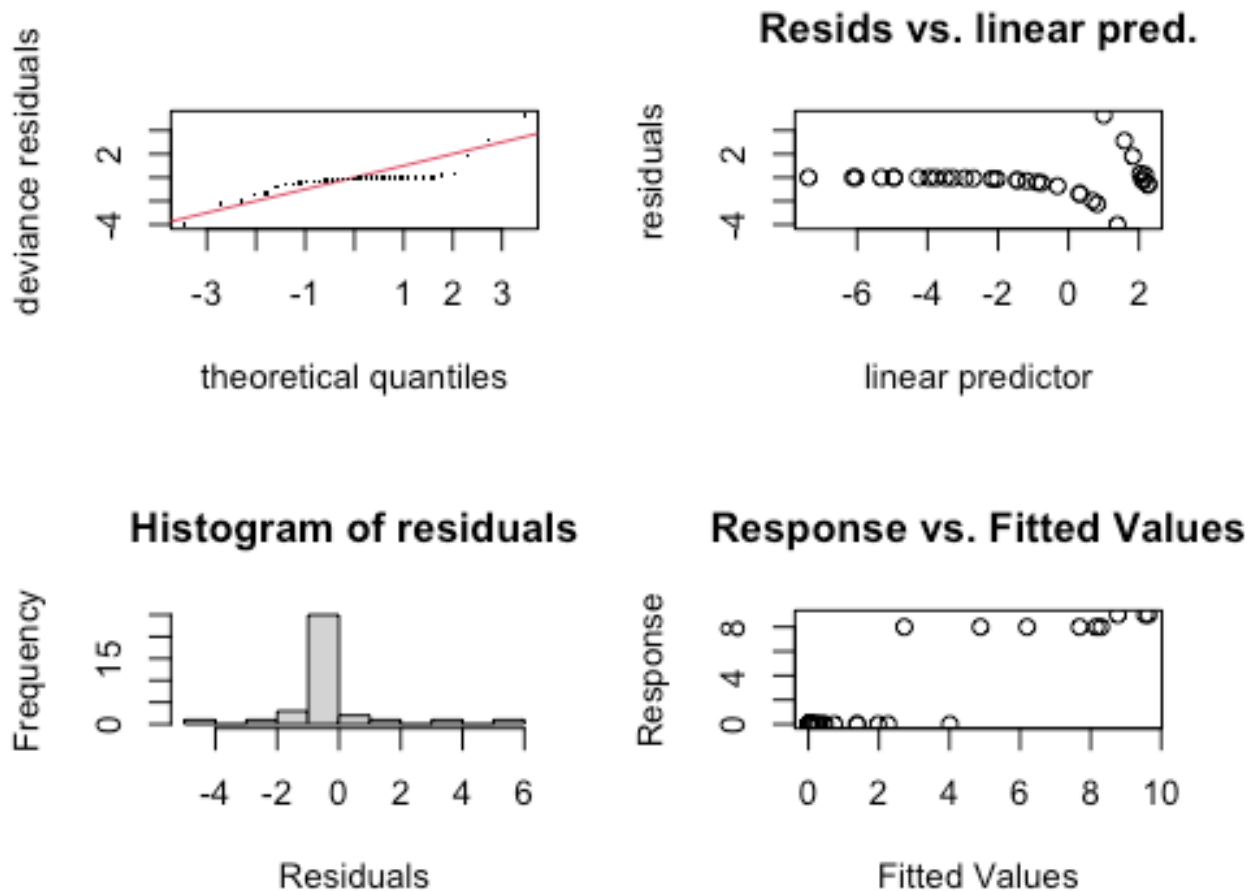
Method: REML Optimizer: outer newton full convergence after 12 iterations. Gradient range

[-1.964407e-05,6.40554e-06] (score 55.32928 & scale 3.571869). Hessian positive definite, eigenvalue range [1.200439e-05,10.5307]. Model rank = 9 / 9

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value
s(LightLevel_ue)	2.00	1.94	1.23	0.89
s(Temp_C)	2.00	1.89	1.37	0.99
ti(Temp_C,LightLevel_ue)	4.00	1.00	0.97	0.35

Strain: *Thalassiosira_pseudonana*, GrowthTemp (°C): 22°C

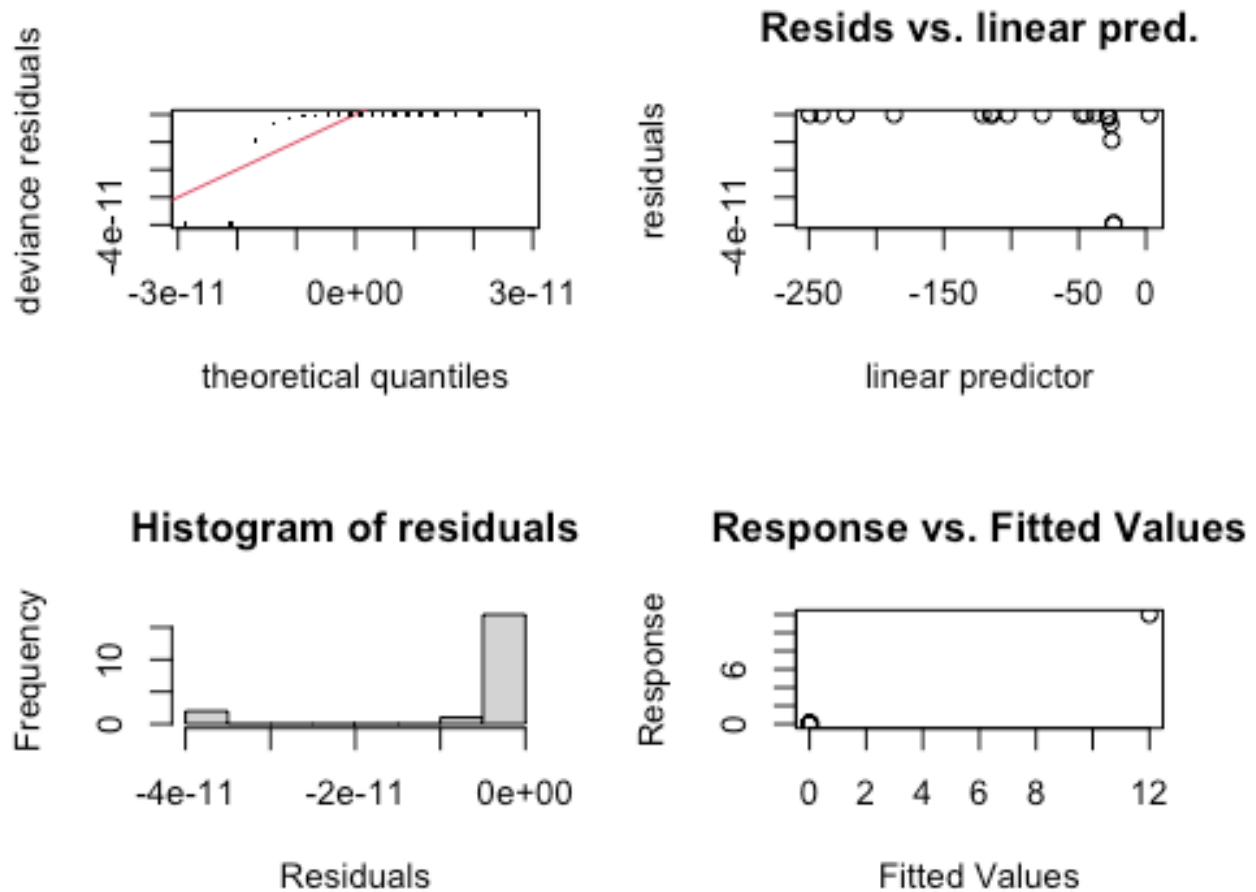


Method: REML Optimizer: outer newton full convergence after 11 iterations. Gradient range [-6.831923e-05,0.0001971058] (score 71.16439 & scale 2.490609). Hessian positive definite, eigenvalue range [5.281984e-06,15.52403]. Model rank = 9 / 9

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value		k'	edf	k-index	p-value
s(LightLevel_ue)	2.00	1.96	1.60	1.00	s(Temp_C)	2.00	1.00	1.02	0.47
ti(Temp_C,LightLevel_ue)	4.00	2.17	1.48	0.99					

Strain: *Chlamydomonas_euryale*, GrowthTemp (°C): 24°C



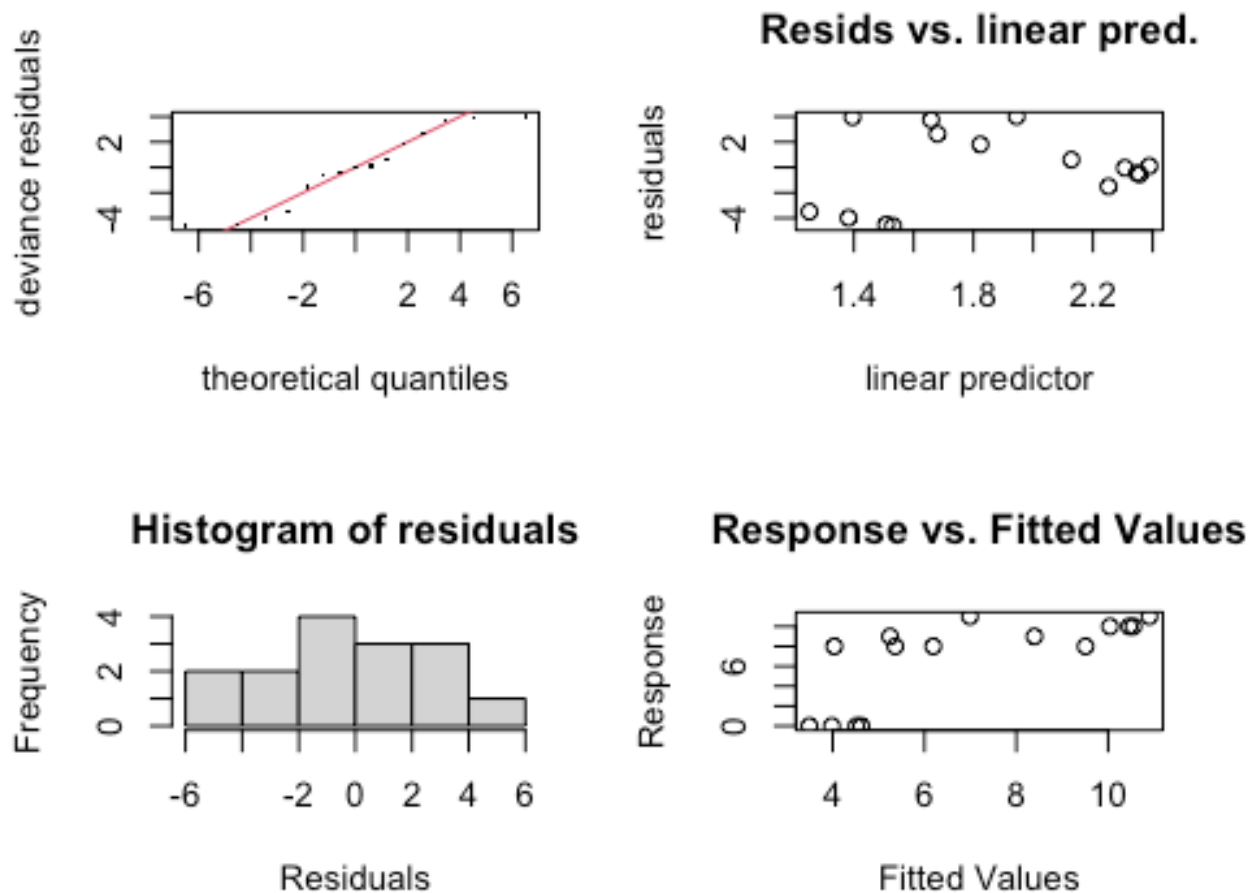
Method: REML Optimizer: outer newton step failed after 1 iteration. Gradient range [-1.61541e-06,9.174523] (score -420.8737 & scale 2.151999e-22). eigenvalue range [-9224757,7.998127]. Model rank = 9 / 9

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

k'	edf	k-index	p-value
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s(LightLevel_ue) 2.00 1.00 0.29 0.015 * s(Temp_C) 2.00 1.96 0.75 0.150
 ti(Temp_C,LightLevel_ue) 4.00 1.00 0.67 0.100 . — Signif. codes: 0 “**0.001**” 0.01 “0.05” 0.1
 “1”

Strain: *Chlamydomonas_ICEMDV*, GrowthTemp (°C): 4°C

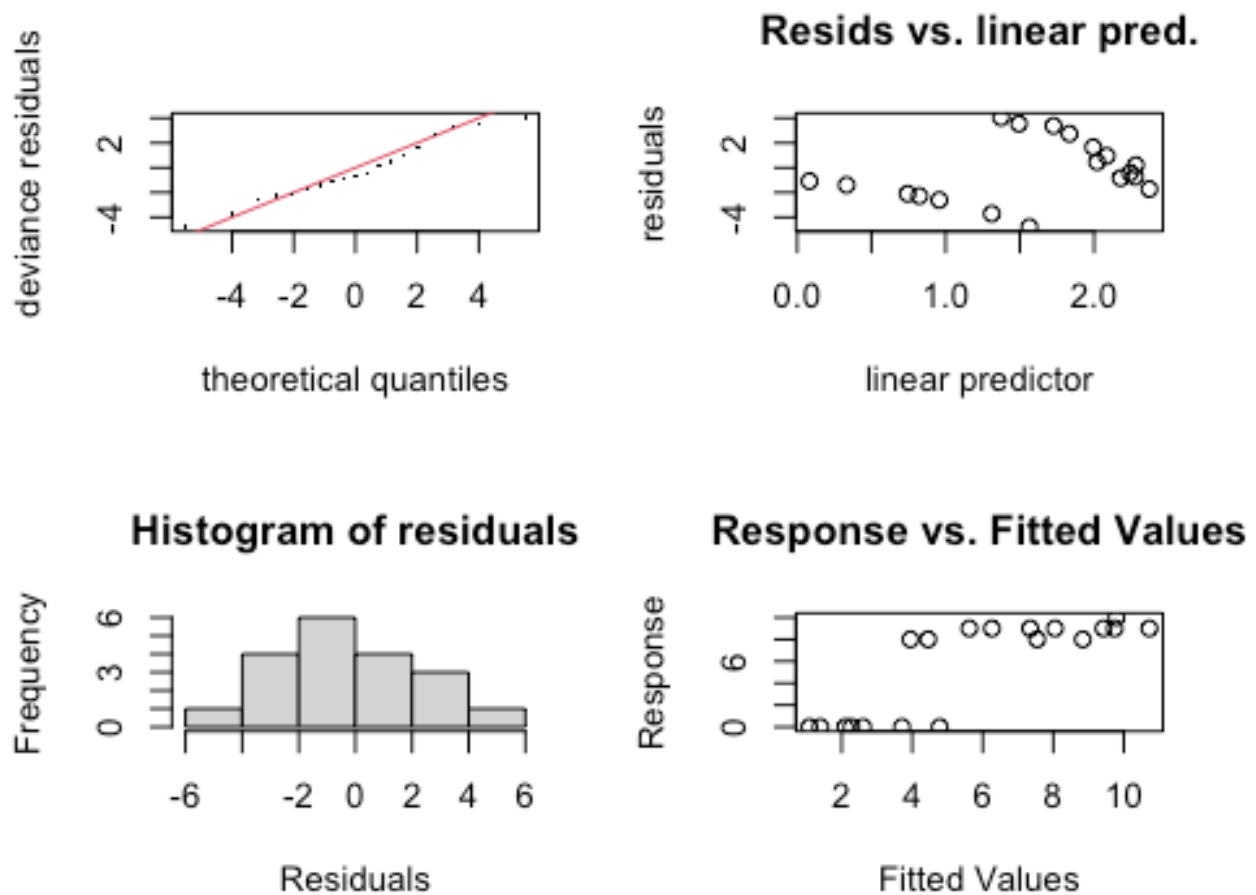


Method: REML Optimizer: outer newton full convergence after 13 iterations. Gradient range
 [-4.513978e-06,3.114671e-06] (score 42.28655 & scale 12.57865). Hessian positive
 definite, eigenvalue range [1.968747e-06,5.520196]. Model rank = 9 / 9

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value
s(LightLevel_ue)	2.00	1.74	1.00	0.47
s(Temp_C)	2.00	1.00	0.99	0.33
ti(Temp_C,LightLevel_ue)	4.00	1.00	1.02	0.35

Strain: *Chlamydomonas_klinobasis*, GrowthTemp (°C): 4°C



Method: REML Optimizer: outer newton full convergence after 10 iterations. Gradient range

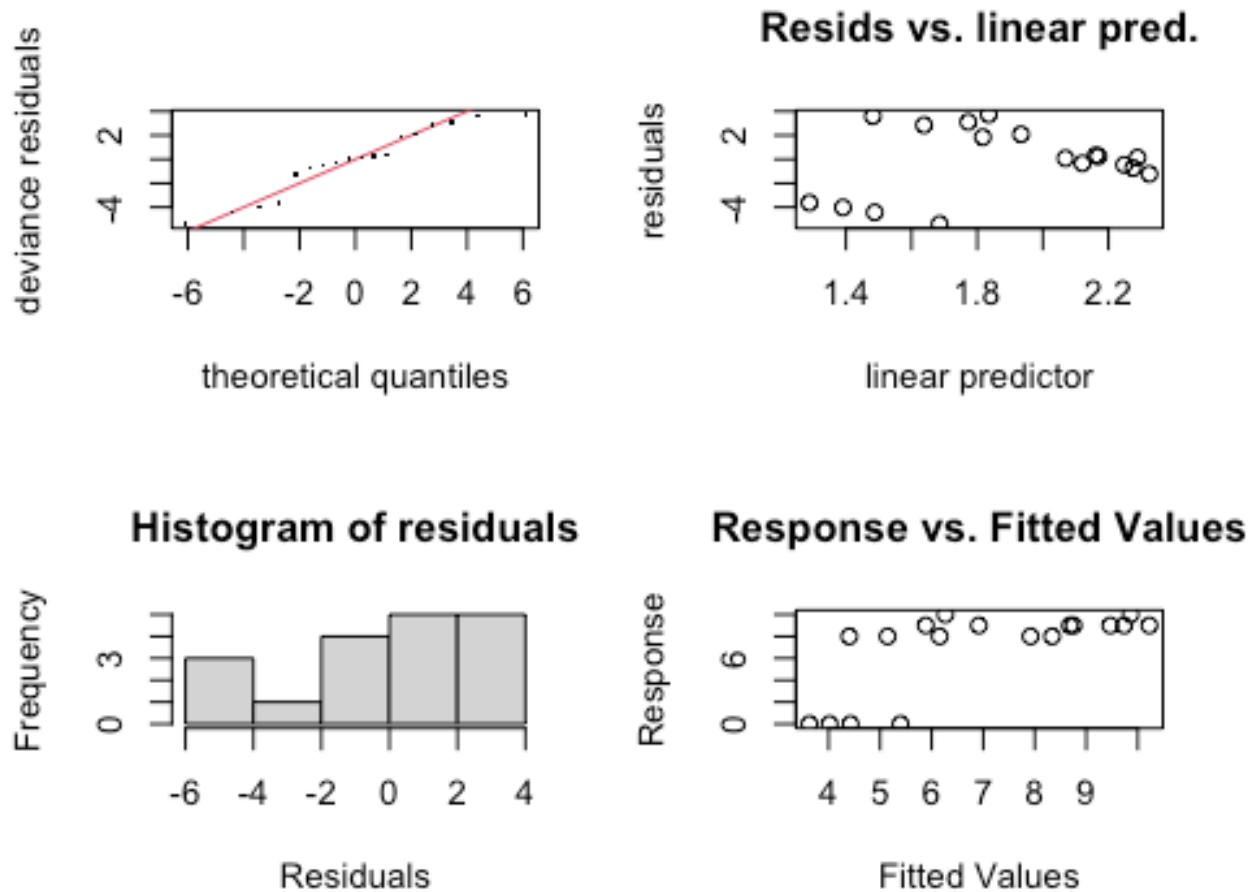
[-1.145248e-05,7.291372e-06] (score 49.70235 & scale 8.081872). Hessian positive definite, eigenvalue range [3.948816e-06,7.521732]. Model rank = 9 / 9

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value
s(LightLevel_ue)	2.00	1.85	0.70	0.085
s(Temp_C)	2.00	1.25	1.14	0.705
ti(Temp_C,LightLevel_ue)	4.00	1.00	0.77	0.080

— Signif. codes: 0 “**0.001**” 0.01 “0.05” 0.1
 ’’ 1

Strain: *Chlamydomonas_malina*, GrowthTemp (°C): 4 °C



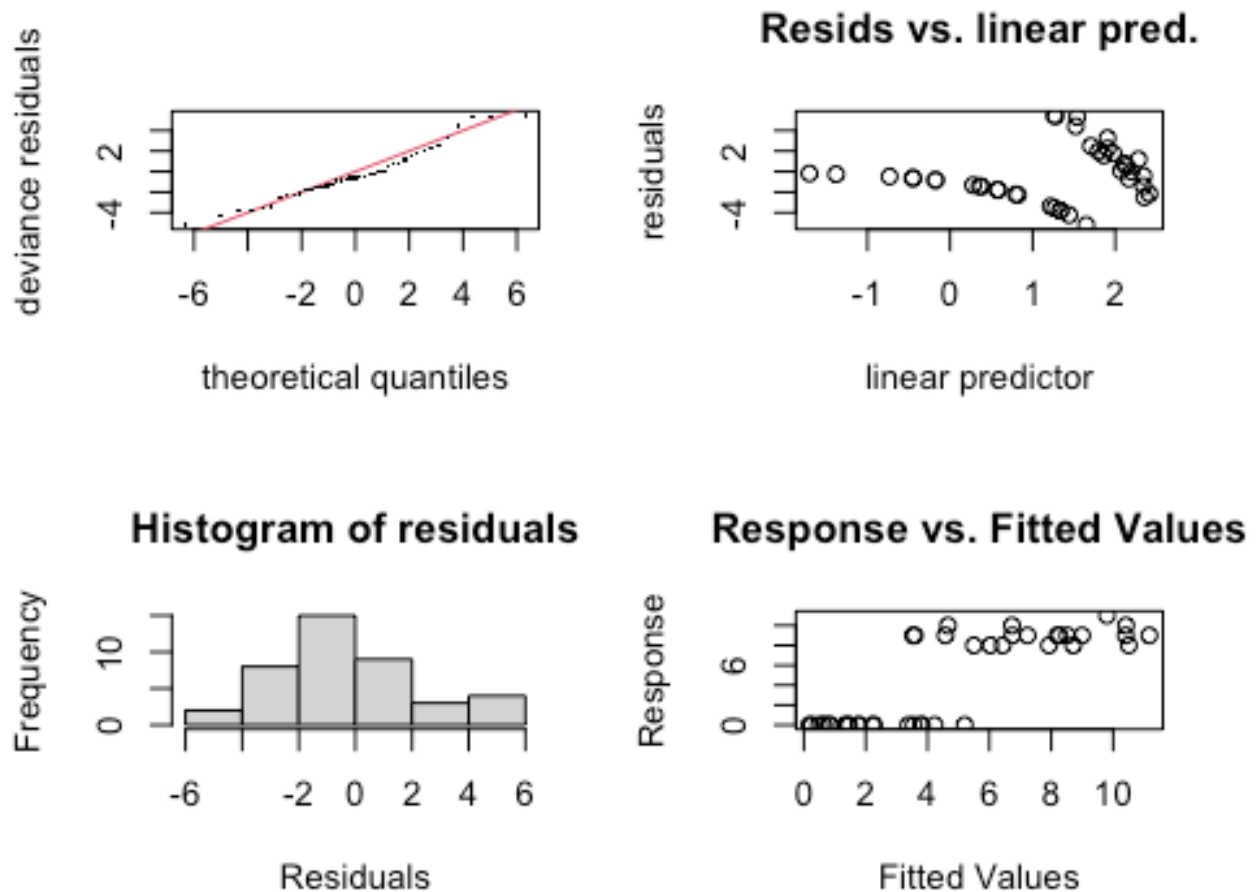
Method: REML Optimizer: outer newton full convergence after 8 iterations. Gradient range [-2.055142e-05,1.125567e-05] (score 49.36194 & scale 10.11107). Hessian positive definite, eigenvalue range [9.922912e-07,7.021343]. Model rank = 9 / 9

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

k'	edf	k-index	p-value
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s(LightLevel_ue) 2.00 1.84 1.04 0.43 s(Temp_C) 2.00 1.00 1.06 0.52
 ti(Temp_C,LightLevel_ue) 4.00 1.00 1.04 0.43

Strain: *Chlamydomonas_priscui*, GrowthTemp (°C): 4°C

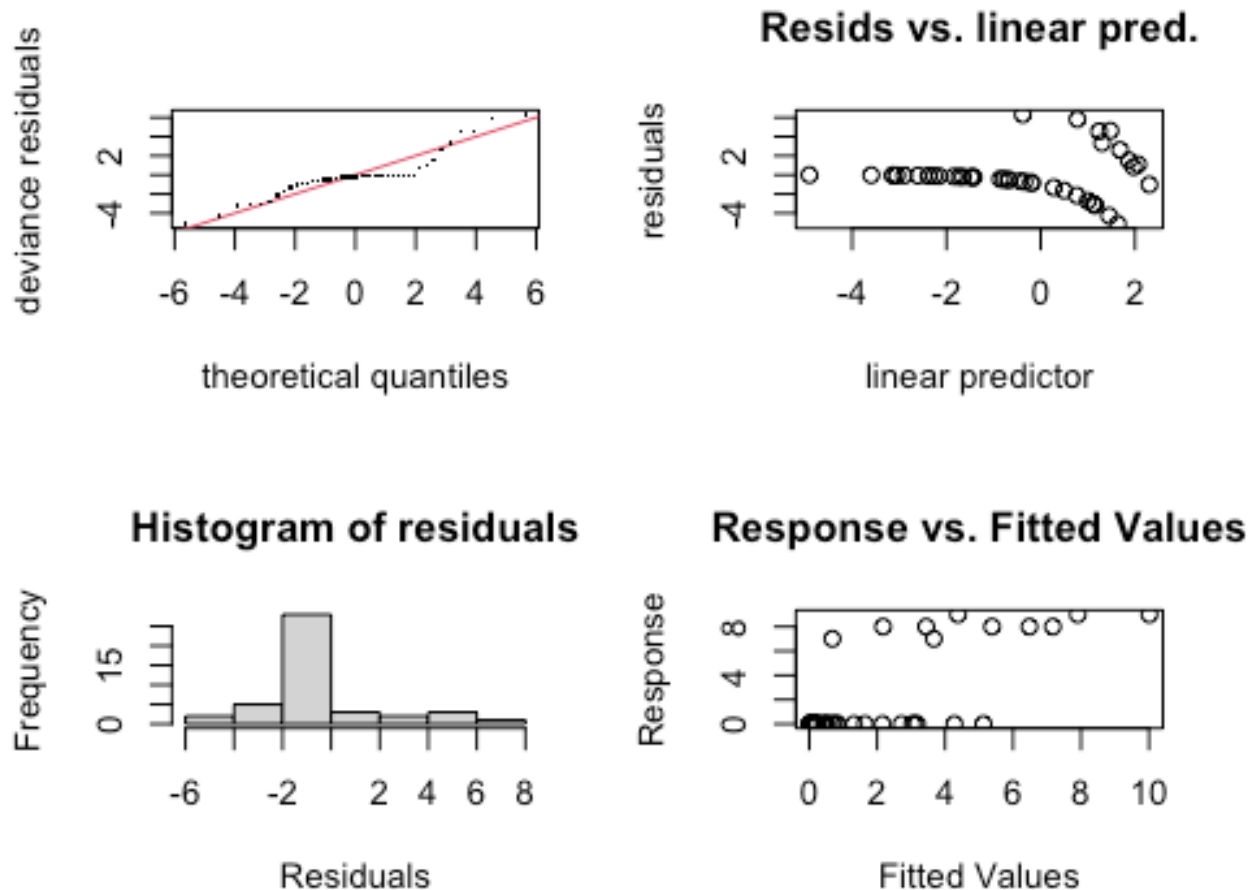


Method: REML Optimizer: outer newton full convergence after 12 iterations. Gradient range [-3.720374e-05,8.668272e-05] (score 104.6995 & scale 7.872017). Hessian positive definite, eigenvalue range [2.611616e-05,18.51441]. Model rank = 9 / 9

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value
s(LightLevel_ue)	2.00	1.90	0.72	0.035 *
s(Temp_C)	2.00	1.77	1.06	0.645
ti(Temp_C,LightLevel_ue)	4.00	1.00	0.62	<2e-16 *** — Signif. codes: 0 " 0.001 " 0.01 " 0.05 ? 0.1 " 1

Strain: *Chlamydomonas_reinhardtii*, GrowthTemp (°C): 24°C



Method: REML Optimizer: outer newton full convergence after 9 iterations. Gradient range [-6.765617e-05,3.899604e-05] (score 103.853 & scale 6.13007). Hessian positive definite, eigenvalue range [5.51875e-05,20.0096]. Model rank = 9 / 9

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value
s(LightLevel_ue)	2.00	1.87	1.01	0.42
s(Temp_C)	2.00	1.00	1.27	0.95
ti(Temp_C,LightLevel_ue)	4.00	2.19	0.85	0.05 *
Signif. codes: 0 '0.001' '0.01' '0.05' '0.1' '1'				

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